PROTEIN ENGINEERING

BONDS, ENERGIES, BUILDING BLOCKS OF PROTEINS

Several atoms are assembled and held together to form thousands of molecules which participate in the building and function of physical and biological systems. A **bond** is any force which holds two atoms together. The formation of bond between two atoms is due to some redistribution or regrouping of electrons to form a more stable configuration. The regrouping of electrons in the combining atoms may take place in either of the 3 ways :

(*a*) by a *transfer* of one or more electrons from one atom to another— **electrovalent bonding**

(*b*) by a *sharing* of one or more pairs of electrons between the combining atoms— **covalent bonding**

(*c*) by a combination of the two processes of *transfer and sharing*— **coordinate bonding.**

Electrovalent or Polar or Ionic Bond

Ionic bond formation takes place between atoms of strongly electropositive and strongly electronegative elements. An element preceding an inert gas in the periodic table is strongly electronegative and the element immediately following the inert gas is strongly electropositive. For example, chlorine is electronegative while sodium is electropositive. According to W. Kossel (1916), a transfer of electron(s) takes place from the outermost shell of the electropositive atom to the outermost shell of the electronegative atom, resulting in the formation of stable positive and negative ions respectively which are held together by electrostatic forces of attraction to form a *molecule* or more precisely an *ion pair*. Thus, The atoms involved are electrically neutral before combining. The element A which has lost its electrons is known as electropositive whereas element B which has gained the electrons is termed electronegative element. The compound formed by electron transfer is termed as *electrovalent* by Langmuir (1919) because the resulting compound is electrolyte. It is also called *polar* since the molecule develops a positive and a negative pole. The electrovalent compounds always exist in ionic form, are hard and nonvolatile, have high melting and boiling points because of stronger nature of the bond and are soluble in polar solvents (such as water and alcohol) and because of the presence of ions conduct electricity in solution or in the fused state. The electrovalent compounds having identical electronic configuration exhibit the phenomenon of isomorphism. For example, NaF (2, 8), magnesium oxide (2, 8), calcium chloride (2, 8, 8) and potassium sulfide (2, 8, 8) are isomorphous.

Electrovalent linkage is common in inorganic compounds like NaCl (Fig. 4-2), Na2O, CaO etc.

The number of the charge (+ or −) or the number of electrons transferred during an ionic bond formation is called the *electrovalence number* or the *electrovalency* of the participating atoms. Thus, sodium has an electrovalency of 1 while calcium or oxygen has 2. The sign (+ or −) of this

number depends upon whether the electron is donated or received. The element donating electron has a positive (+) electrovalency while that receiving electron has a negative (−) sign. Since the number of electrons transferred from an electropositive element to an elecronegative element during an ionic bond formation is equal to the number of electrons present in its outermost orbit, the electrovalency for such elements is the number of electrons present in outermost orbit. Similarly, the electrovalency of an electronegative element is the number of electrons which it can accept to achieve an inert gas configuration.

Covalent or Nonpolar Bond

Covalent bond formation, first suggested by **G.N. Lewis (1916)**, consists in sharing or holding a pair of electrons in partnership between two combining atoms, so that the pair counts towards the electronic grouping of both atoms.

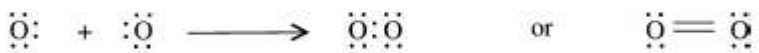
By this mechanism also, the stability akin to the inert gas is attained by each atom. For each pair of electrons to be shared between two atoms, each of the constituent atom contributes one electron : This type of linkage which is the result of equal contribution and equal sharing of electrons is known as covalent bond. The compound formed by electron sharing is termed as *covalent* or *nonpolar* by Langmuir (1919). The covalent compounds always exist in molecular form, are nonelectrolytes or nonionizable, soluble in organic solvents (such as benzene, ether, pyridine etc) and have low melting and boiling points because of weaker nature of the bond. They are usually liquids or gases and are generally soft, easily-fusible and volatile. They are nonconducting in the fused state or in solution. The covalent bond is rigid and directional and as such there is a possibility of position isomerism and stereoisomerism amongst these compounds. *Covalent linkage is common in organic compounds*, although inorganic compounds also have it.

In covalent compounds, one pair of shared electrons corresponds to a single bond, two pairs to double bond, three pairs of electrons or six shared electrons to triple bond. Some common examples from 3 categories are :

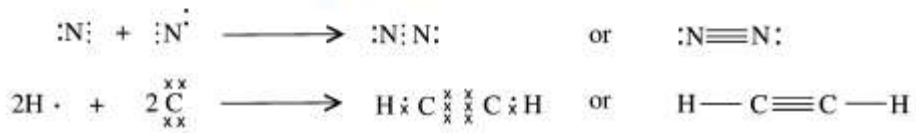
(*a*) One pair of electrons : single bond



(*b*) Two pairs of electrons : double bond



(*c*) Three pairs of electrons : triple bond



With the exception of hydrogen atom which by sharing one electron easily attains the helium configuration, it may be observed that only negative atoms can form covalent bonds by sharing electrons between them. Positive atoms contain an excess of electrons, which they have to lose to attain a stable inert gas structure; and therefore they form electrovalent bonds. The force of bond formation in covalent bonds is the same as that in ionic bonds, *i.e.*, electrostatic attraction between the two atoms, although this force of attraction develops in a different manner. When two atoms, destined to link through a covalent bond, come within a definite range, the wave function of the electrons of two atoms overlap each other. This overlapping causes the accumulation of the negative charge between the two atomic nuclei. The accumulated negative charge, in turn, attracts the nuclei (with positive charges) of the two atoms and, thus, they are held together.

Polar and nonpolar covalent bonds

Polar bond. Polar covalent bonds may be characterized as transition state between ionic and covalent bonds. In this case, there is neither the complete transfer of electrons from one atom to the other nor equal sharing. When a covalent bond is formed between two atoms with *different electronegativities*, the electrons involved in the bond are not shared equally. The atom with higher electronegativity pulls the bonding electrons closer to it. In other words, the electron density in the molecular orbital would be greater around the atom with higher electronegativity. The result of this displacement of the molecular orbital toward the more electronegative atom will acquire a small negative charge and the less electronegative atom will acquire a small positive charge. For example, in the case of **C—Cl bond**, chlorine is more electronegative than carbon. As such, the electron density in the molecular orbital would be higher around chlorine atom than around the carbon atom. Thus, the chlorine atom acquires a small negative charge and the carbon atom acquires a small positive charge. Such a bond which appears to have a positive end and a negative end is said to be a **polar bond** Water molecule is another good example of a polar bond. In this molecule, sharing of electrons between one oxygen and two hydrogen atoms is not equal. The electrons of hydrogen atoms are attracted more towards the nucleus of the oxygen atom. Consequently, oxygen develops a $\delta-$ while

Coordinate or Semipolar or Dative Bond

In a purely covalent bond, each of the electrons in a shared pair or duplet is contributed by each of the two combining atoms; in other words, we have a sharing of give-and-take character. Equal contribution and equal sharing of electrons is thus characteristic of polar bonding. Coordinate bond is also formed by mutual sharing of electrons but in this case the two electrons that are shared come from the same atom (Perkins, 1921). The shared pair of electrons is called *lone pair*. The atom which provides the pair of electrons is called the **donor** and the atom accepting this pair is called the **acceptor**. After the formation of the bond, the lone pair of electrons is held in common. This sort of bonding is called **coordinate** (Sidgwick) or **dative** (Menzies). In this mechanism, although the sharing is equitable, the contribution is one-sided, and therefore a slight polarity develops in the molecule. For this reason, this bond is also called **semipolar** (Sugden).

This type of linkage is represented by an arrow, pointing away from the donor atom (or pointing toward the acceptor atom). Usually the donor is an atom which has already acquired stable electronic configuration and the acceptor is generally two short of the stable configuration.
Coordinate compounds exhibit characteristics similar to covalent compounds :
1. They do not ionize in water and are poor conductors of electricity.
2. They are very sparingly soluble in water but dissolve readily in organic slovents.
3. Since a coordinate linkage is semipolar, the coordinate compounds possess melting and boiling points which are higher than those of purely covalent compounds but lower than those of ionic compounds.
4. The coordinate linkage is also rigid and directional. Therefore, such compounds exhibit space isomerism.
5. The coordinate linkage is easily broken when donor and acceptor are molecules which are capable of independent existence.

Noncovalent Bonds or Interactions

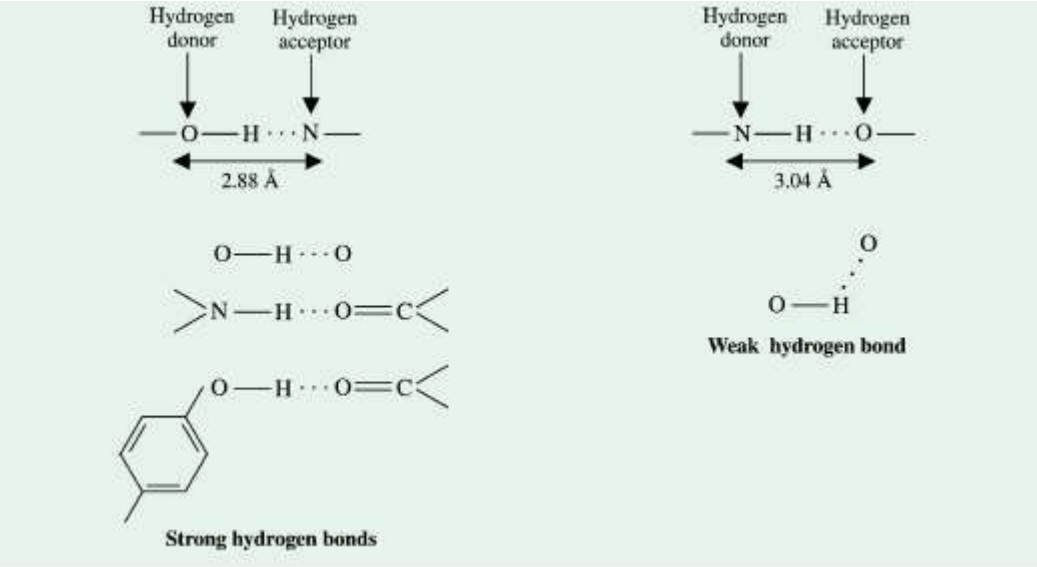
In addition to covalent bonding, there are weaker forces of interaction that profoundly influence conformation of biomolecules and their function. These **noncovalent forces**, as they are called, play key roles in the faithful replication of DNA, the folding of proteins into intricate 3-dimensional forms, the specific recognition of substrates by enzymes, and the detection of signal molecules. *Indeed, all biological structures and processes depend on the interplay of noncovalent interactions as well as covalent ones.* With respect to bonding, weak and strong are used to indicate the amount of energy in a bond. **Strong bonds** such as covalent bonds found in biomolecules require an average of 100 kilocalories/ mole or kcal mol−1, to be cleaved and hence are stable and seldom break under physiological conditions. In contrast, **weak bonds,** such as hydrogen bonds, have energies of 2 to 7 kcal mol−1 and are easily broken. Weak bonds are transient; individually they form and break in small fractions of a second. The transient nature of noncovalent interactions confers flexibility on macromolecules, such as proteins and nucleic acids, that is critical to their function. Furthermore, the large number of noncovalent interactions in a single macromolecule makes it unlikely that at any given moment, all the interactions will be broken; thus macromolecular structures are stable over time. The four fundamental noncovalent bonds are : electrostatic (or ionic) bonds, hydrogen bonds, hydrophobic bonds and van der Waals forces or bonds. They differ in geometry, strength and specificity and are profoundly affected by the presence of water.

Electrostatic or ionic bonds

Ionic bonds are formed due to the attraction between atoms or groups, of opposite charges

(+ and −). A charged group on a substrate can attract an oppositely-charged group on an enzyme. The force (*F*) of such an electrostatic attraction is given by Coulomb's law. where, *q*1 and *q*2 = charges of the two atoms or groups, *r* = distance between the two atoms or groups, and *D* = dielectric constant of the medium. The attraction is strongest in a vaccum where *D* is 1 and is weakest in a medium such as water where *D* is 80. The distance between oppositely-charged atoms in an optimal electrostatic attraction is about 2.8 Å. The average bond energy of ionic bonds in aqueous solution is about 5 kcal mol−1. This kind of attraction is also called *saline bond, salt linkage, salt bridge* or *ion pair*.

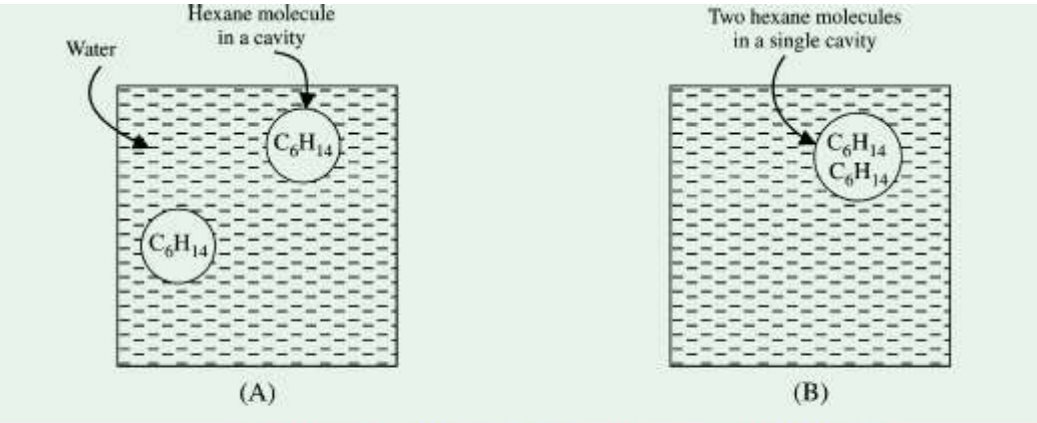
Hydrogen Bonds Hydrogen bonds form between hydrogen atoms covalently bonded to an electronegative atom (such as oxygen or nitrogen) and a second electronegative atom that serves as the hydrogen bond acceptor. Several important biological examples are involved in protein structure formation. Hydrogen bonds, at strength of 12 to 30 kJ/mol, are stronger than van der Waals forces and have an additional property: H bonds tend to be highly directional, forming straight bonds between donor, hydrogen, and acceptor atoms. Hydrogen bonds are also more specific than van der Waals interactions because they require the presence of complementary hydrogen donor and acceptor groups.



Hydrophobic or nonpolar interactions

The essentiality of hydrophilic (water-loving) properties of biomolecules is obvious. That hydrophobic (water-fearing) characteristics can be valuable may not be as readily apparent . As in the formation of micelles, hydrophobic groups of macromolecules, if in proper spatial relation, will interact (*not bond*) to the exclusion of solvent molecules (water) and thereby reside in a hydrophobic environment. On the contrary, hydrophilic groups usually remain exposed to the aqueous environment where they interact with water molecules. Hydrophobic interactions are a

major driving force in the folding of macromolecules, the binding of substrates to enzymes and the formation of membranes that define the boundaries of cells and their internal compartments.
In macromolecules such as proteins, the acceptance or rejection by the aqueous environment of the hydrophilic and hydrophobic moieties respectively exerts a dominant influence on their final conformation. Here the nonpolar side chains of neutral amino acids tend to be closely associated with one another. *The relationship is nonstoichiometric*; hence no true bond may be said to exist. *This clustering together of nonpolar molecules or groups in water is called hydrophobic interaction*. The familiar sight of dispersed oil droplets coming together in water to form a single large oil drop is an analogous process.



Schematic of two molecules of hexane in a small volume of water
The hexane molecules occupy different cavities in the water structure (A), or they occupy the same cavity, which is energetically more favoured (B).

VANDERWALLS INTERACTION:

Van der Waals forces are the result of induced electrical interactions between closely approaching atoms or molecules as their negatively-charged electron clouds fluctuate instantaneously in time. These fluctuations allow attractions to occur between the positively charged nuclei and the electrons of nearby atoms. Van der Waals interactions include dipole–dipole interactions, whose interaction energies decrease as $1/r$ 3; dipole-induced dipole interactions, which fall off as $1/r$ 5; and induced dipole-induced dipole interactions, often called dispersion or London dispersion forces, which diminish as $1/r$ 6. Dispersion forces contribute to the attractive intermolecular forces between all molecules, even those without permanent dipoles, and are thus generally more important than dipole–dipole attractions. Van der Waals attractions operate only over a limited inter atomic distance and are effective bonding interactions at physiological temperatures only when a number of atoms in a molecule can interact with several atoms in a neighboring molecule. For this to occur, the atoms on interacting molecules must pack together neatly. That is, their molecular surfaces must possess a degree of structural complementarity. At best, van der Waals interactions are weak and individually contribute 0.4 to 4.0 kJ/mol of stabilization energy. However, the sum of many such interactions within a macromolecule or between macromolecules can be substantial. For example, model studies of heats of sublimation show that each methylene group in a crystalline hydrocarbon accounts for 8 kJ, and each COH group in a benzene crystal contributes 7 kJ of van der Waals

energy per mole. Calculations indicate that the attractive van der Waals energy between the enzyme lysozyme and a sugar substrate that it binds is about 60 kJ/mol.

ELECTROMAGNETIC RADIATION

Electromagnetic radiation (**EM radiation** or **EMR**) is a fundamental phenomenon of electromagnetism, behaving as waves propagating through space, and also as photon particles traveling through space, carrying radiant energy. EMR carries energy—sometimes called radiant energy—through space continuously away from the source. EMR also carries both momentum and angular momentum. These properties may all be imparted to matter with which it interacts. The electromagnetic spectrum, in order of increasing frequency and decreasing wavelength, can be divided, for practical engineering purposes, into radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, X-rays and gamma rays.

Radio wave:

Radio waves are a type of electromagnetic radiation with wavelengths in the electromagnetic spectrum longer than infrared light. Radio waves have frequencies from 300 GHz to as low as 3 kHz, and corresponding wavelengths ranging from 1 millimeter (0.039 in) to 100 kilometers (62 mi). Like all other electromagnetic waves, they travel at the speed of light. Naturally occurring radio waves are made by lightning, or by astronomical objects. Radio frequency (RF) energy has been used in medical treatments for over 75 years, generally for minimally invasive surgeries and coagulation, including the treatment of sleep apnea. Magnetic resonance imaging (MRI) uses radio frequency waves to generate images of the human body.

Microwaves

Microwaves are a form of electromagnetic radiation with wavelengths ranging from as long as one meter to as short as one millimeter, or equivalently, with frequencies between 300 MHz (0.3 GHz) and 300 GHz. Microwave radiation is used in electron paramagnetic resonance (EPR or ESR) spectroscopy, typically in the X-band region (~9 GHz) in conjunction typically with

magnetic fields of 0.3 T. This technique provides information on unpaired electrons in chemical systems, such as free radicals or transition metal ions such as Cu(II). Microwave radiation is also used to perform rotational spectroscopy and can be combined with electrochemistry as in microwave enhanced electrochemistry. Microwaves do not contain sufficient energy to chemically change substances by ionization, and so are an example of non-ionizing radiation. The word "radiation" refers to energy radiating from a source and not to radioactivity. It has not been shown conclusively that microwaves (or other non-ionizing electromagnetic radiation) have significant adverse biological effects at low levels. Some, but not all, studies suggest that long-term exposure may have a carcinogenic effect. When injury from exposure to microwaves occurs, it usually results from dielectric heating induced in the body. Exposure to microwave radiation can produce cataracts by this mechanism, because the microwave heating denatures proteins in the crystalline lens of the eye (in the same way that heat turns egg whites white and opaque). The lens and cornea of the eye are especially vulnerable because they contain no blood vessels that can carry away heat.

Infrared

Infrared (**IR**) is electromagnetic radiation with longer wavelengths than those of visible light, extending from the nominal red edge of the visible spectrum at 700 nanometers (nm) to 1 mm. This range of wavelengths corresponds to a frequency range of approximately 430 THz down to 300 GHz. Infrared vibrational spectroscopy (see also near-infrared spectroscopy) is a technique that can be used to identify molecules by analysis of their constituent bonds. Each chemical bond in a molecule vibrates at a frequency characteristic of that bond. A group of atoms in a molecule (e.g., CH_2) may have multiple modes of oscillation caused by the stretching and bending motions of the group as a whole. If an oscillation leads to a change in dipole in the molecule then it will absorb a photon that has the same frequency. The vibrational frequencies of most molecules correspond to the frequencies of infrared light. Typically, the technique is used to study organic compounds using light radiation from 4000–400 cm^{-1}, the mid-infrared. A spectrum of all the frequencies of absorption in a sample is recorded. This can be used to gain information about the sample composition in terms of chemical groups present and also its purity.

Visible spectrum:

The **visible spectrum** is the portion of the electromagnetic spectrum that is visible to (can be detected by) the human eye. Electromagnetic radiation in this range of wavelengths is called **visible light** or simply light. A typical human eye will respond to wavelengths from about 390 to 700 nm. astronomical spectroscopy uses high-dispersion diffraction gratings to observe spectra at very high spectral resolutions. Helium was first detected by analysis of the spectrum of the sun. Chemical elements can be detected in astronomical objects by emission lines and absorption lines.

Ultraviolet:

Ultraviolet (**UV**) light is electromagnetic radiation with a wavelength shorter than that of visible light, but longer than X-rays, that is, in the range between ~310 nm and 100 nm, corresponding to photon energies from 4 eV to 12.4 eV. It is so-named because the spectrum consists of electromagnetic waves with frequencies higher than those that humans identify as the color violet. UV/VIS spectroscopy is widely used as a technique in chemistry to analyze chemical structure, the most notable one being conjugated systems. UV radiation is often used to excite a given sample where the fluorescent emission is measured with a spectrofluorometer. In biological research, UV light is used for quantification of nucleic acids or proteins.

Ultraviolet lamps are used to sterilize workspaces and tools used in biology laboratories and medical facilities. Commercially available low-pressure mercury-vapor lamps emit about 86% of their light at 254 nanometers (nm). UV radiation can be an effective viricide and bactericide. Disinfection using UV radiation is commonly used in wastewater treatment applications and is finding an increased usage in drinking water treatment. Many bottlers of spring water use UV disinfection equipment to sterilize their water. Solar water disinfection is the process of using PET bottles and sunlight to disinfect water. Ultraviolet germicidal irradiation is the generic process to inactivate microorganisms in water, air, medical environments, etc.

X-ray

X-radiation (composed of **X-rays**) is a form of electromagnetic radiation. Most X-rays have a wavelength in the range of 0.01 to 10 nanometers, corresponding to frequencies in the range 30 petahertz to 30 exahertz (3×10^{16} Hz to 3×10^{19} Hz) and energies in the range 100 eV to 100 keV. X-ray wavelengths are shorter than those of UV rays and typically longer than those of gamma rays. X-rays interact with matter in three main ways, through photoabsorption, Compton scattering, and Rayleigh scattering. The strength of these interactions depend on the energy of the X-rays and the elemental composition of the material, but not much on chemical properties since the X-ray photon energy is much higher than chemical binding energies. Photoabsorption or photoelectric absorption is the dominant interaction mechanism in the soft X-ray regime and for the lower hard X-ray energies.

METHODS TO DETERMINE PROTEIN STRUCTURE

Protein nuclear magnetic resonance spectroscopy (usually abbreviated **protein NMR**) is a field of structural biology in which NMR spectroscopy is used to obtain information about the structure and dynamics of proteins. The field was pioneered by, among others, Kurt Wüthrich, who shared the Nobel Prize in Chemistry in 2002. Protein NMR techniques are continually being used and improved in both academia and the biotech industry. Structure determination by NMR spectroscopy usually consists of several following phases, each using a separate set of highly specialized techniques. The sample is prepared, resonances are assigned, restraints are generated and a structure is calculated and validated.

Protein nuclear magnetic resonance is performed on aqueous samples of highly purified protein. Usually the sample consist of between 300 and 600 microlitres with a protein concentration in the range 0.1 – 3 millimolar. The source of the protein can be either natural or produced in an expression system using recombinant DNA techniques through genetic engineering. Recombinantly expressed proteins are usually easier to produce in sufficient quantity, and makes isotopic labelling possible.

The most abundant isotopes of carbon and oxygen, carbon-12 and oxygen-16, have no net nuclear spin, which is the physical property nuclear magnetic resonance spectroscopy exploits. The most abundant isotopes of nitrogen, nitrogen-14, has a net nuclear spin; however, it also has a large quadrupolar moment, a property of the atomic nuclei which prevents high-resolution information to be obtained from this isotope. Thus nuclear magnetic resonance of proteins from natural sources are restricted to utilizing nuclear magnetic resonance based solely on protons. However the less common isotopes, carbon-13 and nitrogen-15, are suitable for nuclear magnetic resonance, and therefore labelling the proteins with these compounds open up possibilities for doing more advanced experiments which also detect or use these nuclei. Isotopical labelling is done by growing the expression host in a growth media enriched with the desired isotopes. Since isotopically enriched compounds remain expensive, organisms are used which are capable of

growing on a defined minimal medium, containing only one carbon-13 source, usually glucose, but occasionally glycerol or methanol, and one nitrogen-15 source such as ammonium chloride or ammonium sulfate . These organisms include organisms such as *Escherichia coli*, which is the most frequently used bacteria, and *Pichia pastoris*, which is the most frequently used yeast.The purified protein is usually dissolved in a buffer solution and adjusted to the desired solvent conditions.

Data collection

Protein NMR utilizes multidimensional nuclear magnetic resonance experiments to obtain information about the protein. Ideally, each distinct nucleus in the molecule experiences a distinct chemical environment and thus has a distinct chemical shift by which it can be recognized. However, in large molecules such as proteins the number of resonances can typically be several thousand and a one-dimensional spectrum inevitably has incidental overlaps. Therefore multidimensional experiments are performed which correlate the frequencies of distinct nuclei. The additional dimensions decrease the chance of overlap and have a larger information content since they correlate signals from nuclei within a specific part of the molecule. Magnetization is transferred between nuclei using radiofrequency pulses and delays via one of many elaborate pulse sequences which allow connections between different nuclei to be detected. The array of nuclear magnetic resonance experiments used on proteins fall in two main categories — one where magnetization is transferred through the chemical bonds, and one where the transfer is through space, irrespective of the bonding structure. The first category is used to assign the different chemical shifts to a specific nucleus, and the second is primarily used to generate the distance restraints used in the structure calculation, and in the assignment with unlabelled protein.

Depending on the concentration of the sample, on the magnetic field of the spectrometer, and on the type of experiment, a single multidimensional nuclear magnetic resonance experiment on a protein sample may take hours or even several days to obtain suitable signal-to-noise ratio through signal averaging, and to allow for sufficient evolution of magnetization transfer through the various dimensions of the experiment. Other things being equal, higher-dimensional experiments will take longer than lower-dimensional experiments.

Typically the first experiment to be measured with an isotope-labelled protein is a 2D heteronuclear single quantum correlation (HSQC) spectrum where "heteronuclear" refers to nuclei other than 1H. In theory the heteronuclear single quantum correlation has one peak for each H bound to a heteronucleus. Thus in the 15N-HSQC one signal is expected for each amino acid residue with the exception of proline which has no amide-hydrogen due to the cyclic nature of its backbone. Tryptophan and certain other residues with N-containing sidechains also give rise to additional signals. The 15N-HSQC is often referred to as the fingerprint of a protein because each protein has a unique pattern of signal positions. Analysis of the 15N-HSQC allows researchers to evaluate whether the expected number of peaks is present and thus to identify possible problems due to multiple conformations or sample heterogeneity. The relatively quick heteronuclear single quantum correlation experiment helps determine the feasibility of doing

subsequent longer, more expensive, and more elaborate experiments. It is not possible to assign peaks to specific atoms from the heteronuclear single quantum correlation alone.

Resonance assignment

In order to analyze the nuclear magnetic resonance data, it is important to get a resonance assignment for the protein. That is to find out which chemical shift in each dimension corresponds to which atom. Several different types of experiments have been invented to achieve this. The procedure depends on whether the protein is isotopically labelled or not, since a lot of the assignment experiments depend on carbon-13 and nitrogen-15.

With unlabelled protein the usual procedure is to record a set of two dimensional homonuclear nuclear magnetic resonance experiments through correlation spectroscopy (COSY), of which several types include conventional correlation spectroscopy, *total correlation* spectroscopy (TOCSY) and nuclear Overhauser effect spectroscopy (NOESY). A two-dimensional nuclear magnetic resonance experiment produces a two-dimensional spectrum. The units of both axes are chemical shifts. The COSY and TOCSY transfer magnetization through the chemical bonds between adjacent protons. The conventional correlation spectroscopy experiment is only able to transfer magnetization between protons on adjacent atoms, whereas in the total correlation spectroscopy experiment the protons are able to relay the magnetization, so it is transferred among all the protons that are connected by adjacent protons. Thus in a conventional correlation spectroscopy, an alpha proton transfers magnetization to the beta protons, the beta protons transfers to the alpha and gamma protons, if any are present, then the gamma proton transfers to the beta and the delta protons, and the process continues. In total correlation spectroscopy, the alpha and all the other protons are able to transfer magnetization to the beta, gamma, delta, epsilon if they are connected by a continuous chain of protons. The continuous chain of protons are the sidechain of the individual amino acids. Thus these two experiments are used to build so called spin systems, that is build a list of resonances of the chemical shift of the peptide proton, the alpha protons and all the protons from each residue's sidechain. Which chemical shifts corresponds to which nuclei in the spin system is determined by the conventional correlation spectroscopy connectivities and the fact that different types of protons have characteristic chemical shifts. To connect the different spinsystems in a sequential order, the nuclear Overhauser effect spectroscopy experiment have to be used. Because this experiment transfers magnetization through space, it will show crosspeaks to all protons that are close in space regardless of whether they are in the same spin system or not. The neighbouring residues are inherently close in space, so the assignments can be made by the peaks in the NOESY with other spin systems.

One important problem using homonuclear nuclear magnetic resonance is overlap between peaks. This occurs when different protons have the same or very similar chemical shifts. This problem becomes greater as the protein becomes larger, so homonuclear nuclear magnetic resonance is usually restricted to small proteins or peptides.

The process of resonance assignment for a nitrogen-15 labelled sample is similar to the homonuclear case. No experiment can be performed that transfers magnetisation between two

spin systems through bonds either. The main difference is the ability to record nitrogen-15 edited three dimensional experiments: TOCSY-N HSQC and NOESY-N-HSQC. These experiments build onto the HSQC experiment, but have an additional proton dimension. It can be visualised as each peak in the HSQC having the TOCSY or NOESY peaks stacked onto it. Thus if the TOCSY peak from an amide proton, HN, has a cross peak to its alpha proton, Halpha, at the coordinates (HN, Halpha) in the TOCSY spectrum, the corresponding peak would be at (HN, Halpha,N) in the TOCSY-N-HSQC. Thus it is possible to resolve overlaps in the proton dimension, if the corresponding nitrogens have chemical shifts distinct from one another.

Restraint generation

In order to make structure calculations a number of experimentially determined restraints have to be generated. These fall into different categories, the most widely used is distance restraints and angle restraints.

Distance restraints

A crosspeak in a NOESY experiment signifies spatial proximity between the two nuclei in question. Thus each peak can be converted in to a maximum distance between the nuclei, usually between 1,8 and 6 angstroms.

The 30 intensity of a noesy peak is proportional to the distance to the minus 6th power, so the distance is determined according to intensity of the peak. The intensity-distance relationship is not exact, so usually a distance range is used.

It is of great importance to assign the noesy peaks to the correct nuclei based on the chemical shifts. If this task is performed manually it is usually very labor intensive, since proteins usually have thousands of noesy peaks. Some computer programs such as CYANA and ARIA/CNS perform this task automatically, coupled to a structure calculation. To obtain as accurate assignments as possible it is a great advantage to have access to carbon-13 and nitrogen-15 noesy experiments, since they help to resolve overlap in the proton dimension. This leads to faster and more reliable assignments, and in turn to better structures.

Angle restraints

In addition to distance restraints, restraints on the torsion angles of the chemical bonds, typically the psi and phi angles can be generated. One approach is to use the Karplus equation, to generate angle restraints from coupling constants. Another approach uses the chemical shifts to generate angle restraints. Both methods use the fact that the geometry around the alpha carbon affects the coupling constants and chemical shifts, so given the coupling constants or the chemical shifts, a qualified guess can be made about the torsion angles.

X-ray crystallography (X-ray diffraction)

X-ray crystallography is the science of determining the arrangement of atoms within a crystal using X-rays scattered from electrons. The oldest and most exact X-ray crystallographic method is *single-crystal X-ray diffraction*; for crystals of sufficient purity and regularity, this method gives the crystal structure, which defines the position of most atoms to within a few tenths of an

Ångström. Since all atoms have electrons, X-ray crystallography may be used on any type of crystal, be they metals or minerals or engineered semiconductors or crystals of biomolecules such as aspirin or proteins. Such crystal structures have many scientific and medical applications; for example, they can account for unusual electronic or elastic properties of a material, shed light on the molecular events of enzyme catalysis or serve as the basis for structure-based drug design. X-ray crystallography requires a crystal to amplify the signal; the periodicity of the electron density is used to diffract the X-rays with manageable measurement error. The technique involves the scattering of X-rays of a single wavelength ("monochromatic X-rays") from a single, pure crystal. This scattering produces a diffraction pattern, a set of intense spots (also called *reflections*) on a screen behind it. The spots can be related to the density of electrons in the crystal through a mathematical operation called a Fourier transform. This technique decomposes the electron density into its spatial frequency components, just as the human ear can discern different musical notes in a chord. Each spot observed on the screen corresponds to a spatial oscillation of the electron density along a particular direction within the crystal**.**

Protein crystallization

Most important part of any protein crystallography studies is protein crystallization. Without perfect crystals of protein (or any other biological samples) it is impossible to carry out any crystallographic structural studies. The aim of protein crystallization is to produce well-ordered protein mono-crystals without any inclusion and large enough to diffract X-Ray beam. Also you can find some techniques for low molecular weight peptide crystallization. Despite very wide knowledge about protein crystallization it is still impossible to predict any conditions for protein crystallization. Requirements for Protein crystallization

- Crystallization by Vapor Diffusion
- Crystallization by Dialysis
- Crystallization in space
- Automated Protein crystallization
- Crystallization conditions

Crystal mounting

Usually, protein crystals are grown in some solutions and before further analysis, protein crystals must be mounted and placed on the detector under the X-Ray beam. Protein Crystallography deals with crystals about 0.1 mm in size, that's why crystal mounting required a very high coordination of all motions and reasonable practice. Best advice is to spend 1-2 days to practice play with cheap protein crystals, for example Lysozyme. Currently a lot of efforts are targeted to automate crystal mounting process, but despite this the manual crystal mounting is a still very important part of protein crystallography. protein crystallography needs Crystal separating

- Crystal mounting in capillary
- Crystal mounting in the loop
- Crystals Freezing/refreezing
- Crystal storage in liquid nitrogen

- Automated crystal mounting
- Dry crystals

Data collection

X-ray Data collection is a central part of all protein crystallography. This chapter describes all modern equipment for X-ray data collection: detectors and X-ray radiation sources. Very important part of X-ray data collection is geometry of experiment. Correct alignment of the crystal sometimes allows improving completeness on the final data set. X-ray data collection is time consuming procedure, and this time is very expensive. It is possible to significantly minimize the time of X-ray data collection by correct prediction of strategy of data collection. Multi-wavelength X-ray data collection, as a part of anomalous data collection will be discussed, as a most popular technique of crystal structure solution in protein crystallography.

Detector types

- Radiation damage
- Strategy prediction
- Anomalous diffraction
- Geometry of the data collection

Data processing

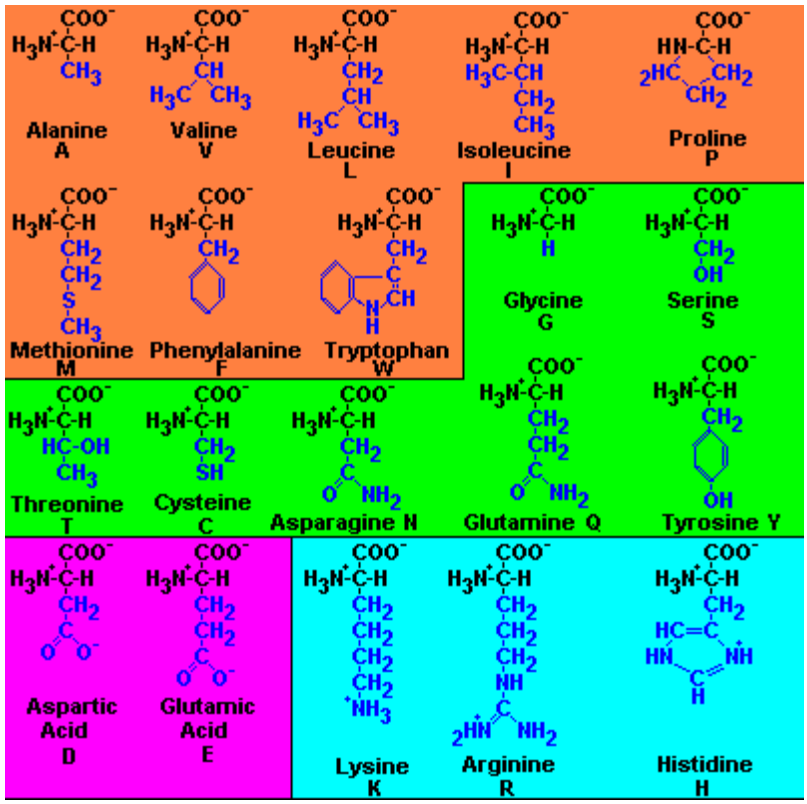
X-ray data processing is a key stage for protein crystallography. The final quality of X-Ray data is related with data processing procedure, which includes integration of the crystallographic data and scaling. In this chapter the basic programs for data processing will be discussed. You also can find manuals for most popular programs written by our specialists with set of samples of data processing for many different cases, from easy one to a very difficult. Also the role of some parameters widely used for data processing, such as integration spot radius or estimated error, will be discussed together with number of examples showing the impact of these parameters of data processing on final model and final statistic.

Amino Acids

Proteins are polymers of amino acids, with each **amino acid residue** joined to its neighbor by a specific type of covalent bond.

20 essential amino acids

Each amino acid contains an "amine" group (**NH3**) and a "carboxy" group (**COOH**) (shown in black in the diagram). The amino acids vary in their side chains (indicated in blue in the diagram). The eight amino acids in the orange area are nonpolar and hydrophobic. The other amino acids are polar and hydrophilic ("water loving"). The two amino acids in the magenta box are acidic ("carboxy" group in the side chain). The three amino acids in the light blue box are basic ("amine" group in the side chain).


Alanine
A
Valine
V
Leucine
L
Isoleucine
I
Proline
P
Methionine
M
Phenylalanine
F
Tryptophan
W
Glycine
G
Serine
S
Threonine
T
Cysteine
C
Asparagine N
Glutamine Q
Tyrosine Y
Aspartic
Acid
D
Glutamic
Acid
E
Lysine
K
Arginine
R
Histidine
H

CHEMICAL REACTIVITY IN RELATION TO POSTTRANSLATIONAL MODIFICATION

In the final stage of protein synthesis, the nascent polypeptide chain is folded and processed into its biologically active form. During or after its synthesis, the polypeptide progressively assumes its native conformation, with the formation of appropriate hydrogen bonds and van der Waals, ionic, and hydrophobic interactions. In this way the linear, or one-dimensional, genetic message in the mRNA is converted into the three- dimensional structure of the protein. Some newly made proteins, both prokaryotic and eukaryotic, do not attain their final biologically active conformation until they have been altered by one or more processing reactions called posttranslational modifications.

Amino-Terminal and Carboxyl-Terminal Modifications the first residue inserted in all polypeptides is N-formylmethionine (in bacteria) or methionine (in eukaryotes). However, the formyl group, the amino-terminal Met residue, and often additional amino-terminal (and, in some cases, carboxyl-terminal) residues may be removed enzymatically in formation of the final functional protein. In as many as 50% of eukaryotic proteins, the amino group of the amino-terminal residue is N-acetylated after translation. Carboxyl-terminal residues are also sometimes modified.

Modification	Description
ACETYLATION	N-terminal of some residues and side chain of lysine
AMIDATION	Generally at the C-terminal of a mature active peptide after oxidative cleavage of last glycine
BLOCKED	Unidentified N- or C-terminal blocking group
FORMYLATION	Generally of the N-terminal methionine
GAMMA-CARBOXYGLUTAMIC ACID	Of glutamate
HYDROXYLATION	Generally of asparagine, aspartate, proline or lysine
METHYLATION	Generally of N-terminal phenylalanine, side chain of lysine, arginine, histidine, asparagine or glutamate, and C-terminal cysteine
PHOSPHORYLATION	Of serine, threonine, tyrosine, aspartate, histidine or cysteine, and, more rarely, of arginine
PYRROLIDONE CARBOXYLIC ACID	N-terminal glutamine which has formed an internal cyclic lactam. This is also called 'pyro-Glu'. Very rarely, pyro-Glu can be produced by modification of a N-terminal glutamate
SULFATION	Of tyrosine, serine or threonine

PROTEIN ARCHITECTURE

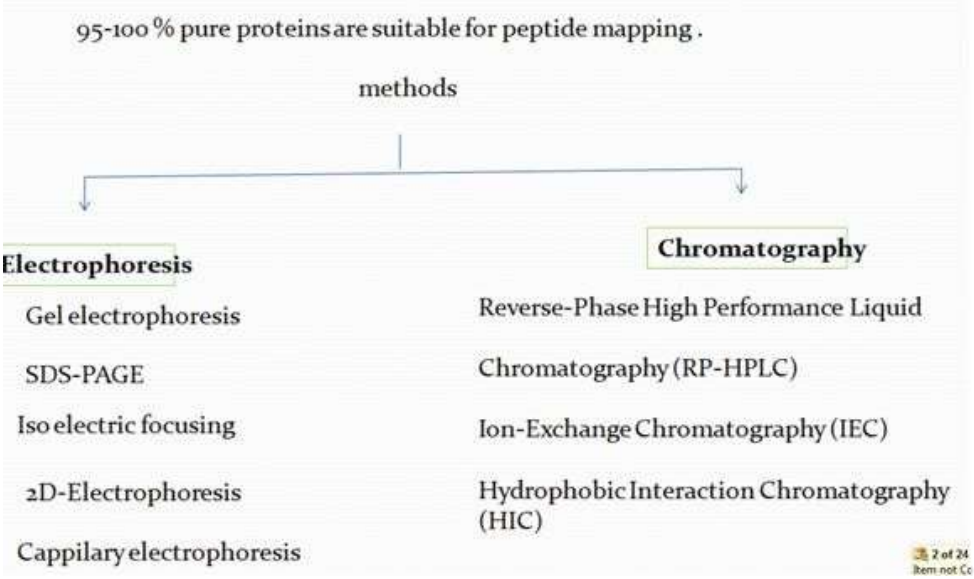
PEPTIDE MAPPING

Peptide mapping is not a general method, but involves developing specific maps for each unique protein. Although the technology is evolving rapidly, there are certain methods that are generally accepted. A peptide map may be viewed as a fingerprint of a protein and is the end product of several chemical processes that provide a comprehensive understanding of the protein being analyzed. Four major steps are necessary for the development of the procedure:

1. Isolation and purification of the protein, if the protein is part of a formulation.
2. Selective cleavage of the peptide bands.
3. Chromatographic separation of the peptides.
4. Analysis and identification of the peptides.

ISOLATION and PURIFICATION:

Isolation and purification are necessary for analysis of bulk drugs or dosage forms containing interfering excipients and carrier proteins and, when required, will be specified in the monograph. Quantitative recovery of protein from the dosage form should be validated.



SELECTIVE CLEAVAGE OF PEPTIDE BONDS

The selection of the approach used for the cleavage of peptide bonds will depend on the protein under test.This selection process involves determination of the type of cleavage to be employed—enzymatic or

chemical—and the type of cleavage agent within the chosen category.Several cleavage agents and their specificity are shown in Table 1.

Table 1.Examples of Cleaving Agents

Type	Agent	Specificity
Enzymatic	Trypsin (EC3.4.21.4)	C-terminal side of Arg and Lys
	Chymrotrypsin (EC3.4.21.1)	C-terminal side of hydrophobic residues (e.g.,Leu,Met,Ala,aromatics)
	Pepsin (EC3.4.23.122)	Nonspecific digest
	Lysyl endopeptidase (Lys-C Endopeptidase)	C-terminal side of Lys
	(EC3.4.21.50)	
	Glutamyl endopeptidase (from S.aureusstain V8)	C-terminal side of Glu and Asp
	(EC23.4.21.19)	
	Peptidyl-Asp metaplo-endopeptidase	N-terminal side of Asp
	(Endoproteinase Asp-N)	
	(EC3.4.24.33)	
	(Clostripan)	C-terminal side of Arg
	(EC3.4.28.8)	
Chemical	Cyanogen bromide	C-terminal side of Met
	2-Nitro-5-thio-cyano-benzoic acid	N-terminal side of Cys
	O-Iodosobenzoic acid	C-terminal side of Trp and Tyr
	Dilute acid	Asp and Pro
	BNPS-skatole	Trp

Pretreatment of Sample:

Depending on the size or the configuration of the protein,different approaches in the pretreatment of samples can be used.For monoclonal antibodies,the heavy and light chains will need to be separated before mapping.If trypsin is used as a cleavage agent for proteins with a molecular mass greater than 100,000Da,lysine residues must be protected by citraconylation or maleylation;otherwise,too many peptides will be generated.

Pretreatment of the Cleavage Agent:

Pretreatment of cleavage agents—especially enzymatic agents—might be necessary for purification purposes to ensure reproducibility of the map.For example,trypsin used as a cleavage agent will have to be treated with tosyl-L-phenylalanine chloromethyl ketone to inactivate chymotrypsin.Other methods,such as purification of trypsin by HPLCor immobilization of enzyme on a gel support,have been successfully used when only a small amount of protein is available.

Pretreatment of the Protein:

Under certain conditions,it might be necessary to concentrate the sample or to separate the protein from added substances and stabilizers used in the formulation of the product,if these interfere with the mapping procedure.Physical procedures used for pretreatment can include ultrafiltration,column chromatography,and lyophilization.

Other pretreatments, such as the addition of chaotropic agents (e.g.,urea),can be used to unfold the protein prior to mapping.To allow the enzyme to have full access to cleavage sites and permit some unfolding of the protein,it is often necessary to reduce and alkylate the disulfide bonds prior to digestion.

Factors that affect the completeness and effectiveness of digestion of proteins:

pH— The pHof the digestion mixture is empirically determined to ensure the optimization of the performance of the given cleavage agent.For example,when using cyanogen bromide as a cleavage agent,a highly acidic environment (e.g.,pH2,formic acid)is necessary;however,when using trypsin as a cleavage agent,a slightly alkaline environment (pH8)is optimal.As a general rule,the pHof the reaction milieu should not alter the chemical integrity of the protein during the digestion and should not change during the course of the fragmentation reaction.

Temperature— Atemperature between 25 $^{\circ}$and 37 $^{\circ}$is adequate for most digestions.The temperature used is intended to minimize chemical side reactions.The type of protein under test will dictate the temperature of the reaction milieu,because some proteins are more susceptible to denaturation as the temperature of the reaction increases.For example,digestion of recombinant bovine somatropin is conducted at 4°,because at higher temperatures it will precipitate during digestion.

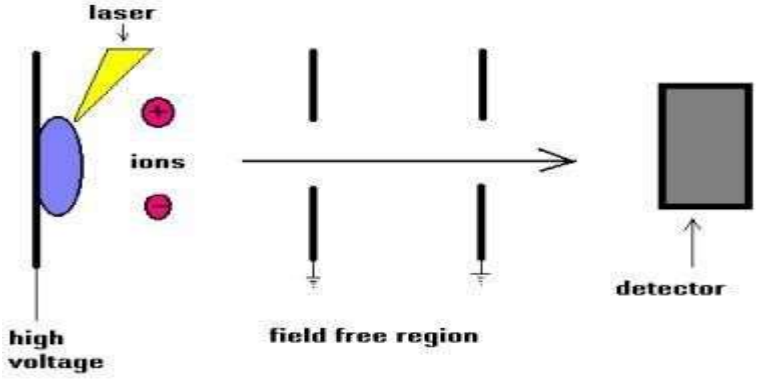
Time— If a sufficient amount of sample is available,a time course study is considered in order to determine the optimum time to obtain a reproducible map and avoid incomplete digestion.Time of digestion varies from 2to 30hours.The reaction is stopped by the addition of an acid that does not interfere in the tryptic map,or by freezing.

Amount of Cleavage Agent— Although excessive amounts of cleavage agent are used to accomplish a reasonably rapid digestion time (i.e.,6to 20hours),the amount of cleavage agent is minimized to avoid its contribution to the chromatographic map pattern.Aprotein to protease ratio between 20:1and 200:1is generally used.It is recommended that the cleavage agent can be added in two or more stages to optimize cleavage.Nonetheless,the final reaction volume remains small enough to facilitate the next step in peptide

mapping—the separation step.To sort out digestion artifacts that might be interfering with the subsequent analysis,a blank determination is performed,using a digestion control with all the reagents,except the test protein.

Methods for protein identification

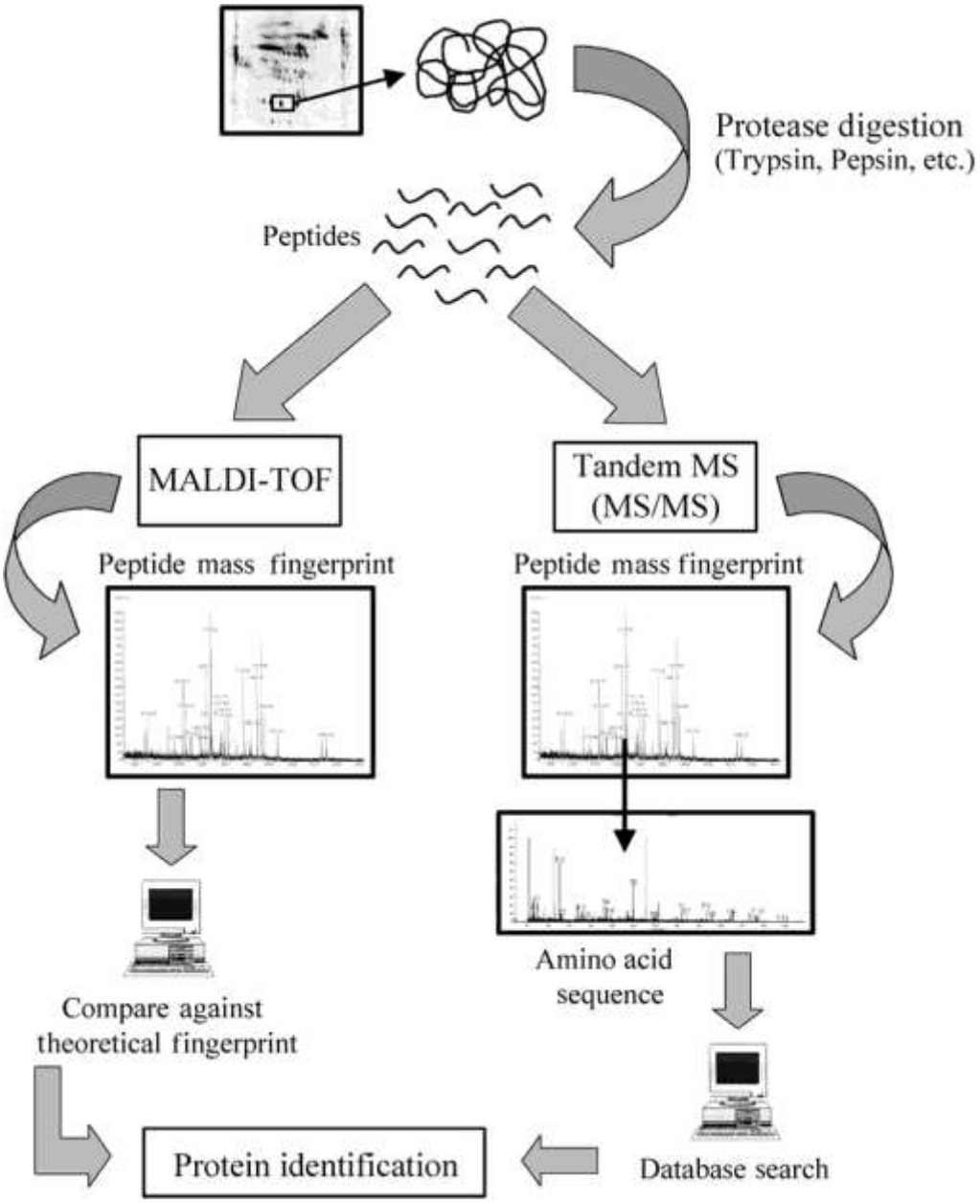
- After protein separation and fractionation, proteins are digested (their internal bonds are cut) by a specific enzyme. The result is a solution of many peptides derived from a single protein
- MALDI-TOF, can produce peptide mass fingerprints.



The main function of the **mass analyser** is to **separate** , or **resolve** , the ions formed in the ionisation source of the mass spectrometer according to their **mass-to-charge (m/z)** ratios. There are a number of mass analysers currently available, the better known of which include **quadrupoles** , **time-of-flight (TOF)** analysers, **magnetic sectors** , and both **Fourier transform** and **quadrupole ion traps** . These mass analysers have different features, including the m/z range that can be covered, the mass accuracy, and the achievable resolution. The compatibility of different analysers with different ionisation methods varies. For example, all of the analysers listed above can be used in conjunction with electrospray ionisation, whereas MALDI is not usually coupled to a quadrupole analyser.
Tandem (MS-MS) mass spectrometers are instruments that have more than one analyser and so can be used for structural and sequencing studies. Two, three and four analysers have all been incorporated into commercially available tandem instruments, and the analysers do not necessarily have to be of the same type, in which case the instrument is a **hybrid** one. More popular tandem mass spectrometers include those of the **quadrupole-quadrupole, magnetic sector-quadrupole** , and more recently, the **quadrupole-time-of-flight** geometries.

- The protein fingerprint is input into the program, and the program determines which sequence in the database would produce the same fingerprint as the one submitted.
  Example:

MASCOT
SWISS-PROT
PepMAPPER



High-Throughput Protein Sequencing

High throughput DNA sequencing revolutionized genomics, which is the study of all the DNA in a living system. The ability to sequence whole genomes has stimulated new disease treatments, increased understanding of microbial diversity and evolutionary origins, and promises to usher in personalized medicine.]. Yet it is ultimately proteins that perform most cellular functions necessary for life. A similar breakthrough in the global study of all proteins (proteomics) requires a corresponding technological advance: high-throughput protein sequencing.

DNA sequencing relies on a molecule called DNA polymerase, a protein that replicates genomes in living systems with the help of associated proteins and co-factors. No obvious 'protein polymerase' candidate presents itself, and current protein sequencing relies on standard biochemical reactions like Edman degradation or mass spectrometry. Furthermore, protein sequencing presents unique challenges. The majority of DNA is spatially localized in cells, and remains at a static concentration when not replicated. In contrast, protein localization, concentration, and expression varies for different cell types, time points, cellular locations and internal environments (pH, temperature, cellular stress). All of these variables present challenges to protein sequencing, yet also highlight the wealth of information that may help explain diseases or track their progression.

One approach for high-throughput protein sequencing involves unfolding proteins into a linear strand and immobilizing them on a substrate with their N-terminals (nitrogen terminals) exposed. One can imagine the N-termini being specifically recognized, coupled to a readout (e.g. an attached fluorescent molecule), and this process repeating iteratively, see Fig. 1.1.

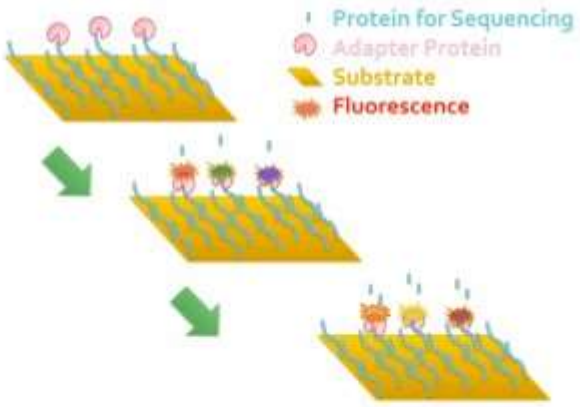


Figure 1.1: N-terminus High-Throughput Protein Sequencing Schema. A protein mixture is immobilized on a matrix and a mixture of adapters are added. These adaptor proteins bind the protein for sequencing depending on its N-terminal amino acid, and have a unique fluorescent molecule attached. The amino acid identity is sequenced by subsequent rounds of cleavage and reapplication of the adapter.

The success of such an approach, among other things, relies on a set of adapter proteins that are able to bind to one amino acid and exclude the other 19 types of amino acids. Also this adapter-peptide binding event needs to be independent of other features of the unfolded peptide, such as the next amino acid in the sequence.

AUTOMATED EDMAN DEGRADATION

This reaction for determining the sequences of peptides and proteins from their N-terminus was reported by Pehr Edman in the early 1950s. It uses a three-stage reaction cycle for step-wise removal of amino acid residues from the N-terminus of a polypeptide chain. The three stages of the Edman cycle

(i) coupling,
(ii) cyclization, and
(iii) conversion.

In the most useful method of N-terminal residue identification, the **Edman degradation** (named after its inventor,Pehr Edman), **phenylisothiocyanate (PITC, Edman's reagent)** reacts with the N-terminal amino groups of proteins under mildly alkaline conditions to form their **phenylthiocarbamyl (PTC)** adduct. This product is treated with an anhydrous strong acid such as trifluoroacetic acid, which cleaves the N-terminal residue as its **thiazolinone** derivative but does not hydrolyze other peptide bonds. *The Edman degradation therefore releases the N-terminal amino acid residue but leaves intact the rest of the polypeptide chain.*The thiazolinone-amino acid is selectively extracted into an organic solvent and is converted to the more stable **phenylthiohydantoin (PTH)** derivative by treatment with aqueous acid.This PTH-amino acid is most commonly identified by comparing its retention time on HPLC with those of known PTH-amino acids. The most important difference between the Edman degradation and other methods of N-terminal residue identification is that *we can determine the amino acid sequence of a polypeptide chain from the N-terminus inward by subjecting the polypeptide to repeated cycles of the Edman degradation and, after every cycle, identifying the newly liberated PTH-amino acid*. This technique has been automated, resulting in great savings of time and materials

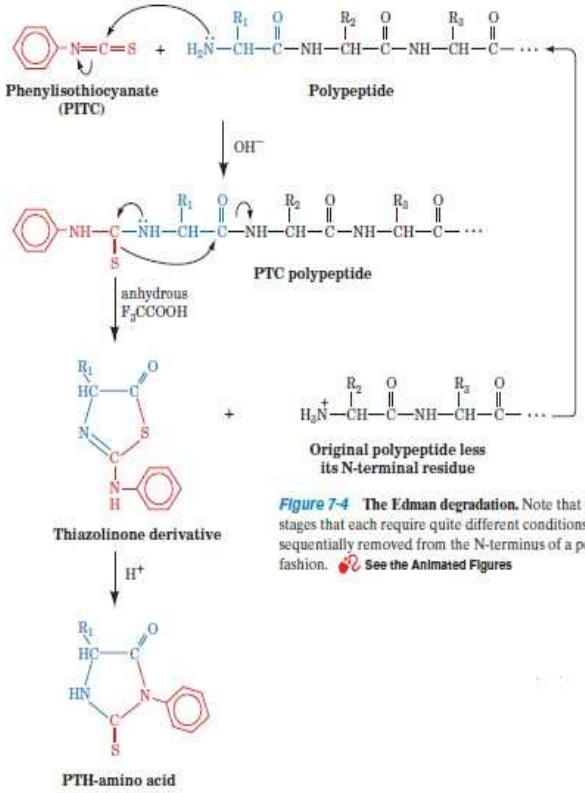


Figure 7-4 **The Edman degradation.** Note that the reaction occurs in three separate stages that each require quite different conditions. Amino acid residues can therefore be sequentially removed from the N-terminus of a polypeptide in a controlled stepwise fashion. See the Animated Figures

In the first-generation sequencers, a "spinning cup" constituted the reaction center, and the sample was kept in place by centrifugal force, making extractions and reactions possible. Extraction losses were a problem in each cycle, however, soon leading Laursen and others to the idea of solid-phase attachment of the peptide to be degraded. Consequently, only a few years later, the solid-phase sequencer was reported. Although both types of sequencers were based on the same Edman chemistry, their different properties, including the different attachments of the samples, made them suitable for different analyses. The liquid-phase sequencer was excellent for proteins, and the solid-phase instrument was better for short peptides, because of their different sensitivities to extractive losses and to build-up of background signals.

Second-Generation Chemical Sequencers: Automation of All Steps

With the development of protein sequence analyzers, it became possible to determine amino acid sequences routinely, and there was an exponential increase in known protein sequences. However, analysis at this stage was still time-consuming and nonautomatic, requiring knowledge and real research. Gradually, however, a set of further inventions increased the automation and brought the actual sequence analysis more or less to the present-day automatic stage. This development essentially relied on three further inventions/improvements.

One concerned the attachment of the protein/peptide to the sequencer for analysis. This important step had several sub-steps. Instead, peptides could now be attached to membranes, essentially moving the degradation from the traditional "liquid phase" system to the advantageous "solid phase" and "gas-phase" systems. The latter are solid phase for the attachment of the peptide to a support, and gas phase for the introduction of some reagents. These approaches are still in use and now allow rapid and sensitive analysis. Many other sub-steps in this transformation were involved. In particular, perhaps the introduction of chemical attachments of peptides to membranes, and the successive development of alternative, miniature column attachments for sample introductions and preparations, should be mentioned.

The second improvement at this stage, increases in speed (because of less washing) and sensitivity (because of higher yield from lack of cross-contamination) made sequencers still more useful and rapid. Recently, this has been carried still further, and soon "chip-based blocks" may be encountered in sequencers, with all solvent delivery and removal stages in extremely small volumes of "chip-blocks".

The third major advance at this stage was the introduction of automatic methods for the conversion of a thiazolinone from each cycle into the corresponding thiohydantoin. This became possible because of the introduction of a second reaction vessel, with separate reagents and reactions.

Once the conversion had become automated, it became possible also to link the subsequent PTH identification step to the cycle of automatic events in the sequencer, thus opening the road to on-line identification of the liberated amino acid derivative in each cycle. These on-line modes were started very early, with the introduction of HPLC, and were soon commercialized and perfected in a new set of complete sequencers, starting with the "gas-phase sequencer" that was available in the 1980s. Soon, the on-line approach was coupled with post-PTH-identification data treatments, allowing extensive computer interpretation at each step. All chromatograms can now be stored and compared with on-line computers and further interpreted and related to sequences in databanks, analyzed in modeling programs, and submitted to further computerized adjustments.

MASS SPECTROSCOPY

Mass spectrometry is a sensitive technique used to detect, identify and quantitate molecules based on their mass and charge (*m/z*) ratio. Originally developed almost 100 years ago to measure elemental atomic weights and the natural abundance of specific isotopes, MS was first used in the biological sciences to trace heavy isotopes through biological systems; later on, MS was used to sequence oligonucleotides and peptides and analyze nucleotide structure.

The development of macromolecule ionization methods, including electrospray ionization (ESI) and matrix-assisted laser desorption/ionization (MALDI), enabled the study of protein structure by MS.

All mass spectrometers have an ion source, a mass analyzer and an ion detector. The nature of these components varies based on the type of mass spectrometer, the type of data required, and the physical properties of the sample. Samples are loaded into the mass spectrometer in liquid, gas or dried form and then vaporized and ionized by the ion source (e.g., APCI, DART, EI).

The charge that these molecules receive allows the mass spectrometer to accelerate the ions throughout the remainder of the system. The ions encounter electric and/or magnetic fields from mass analyzers, which deflect the paths of individual ions based on their mass and charge (*m/z*). Commonly used mass analyzers include time-of-flight [TOF], orbitraps, quadrupoles and ion traps, and each type has specific characteristics. Mass analyzers can be used to separate all analytes in a sample for global analysis, or they can be used like a filter to deflect only specific ions towards the detector.

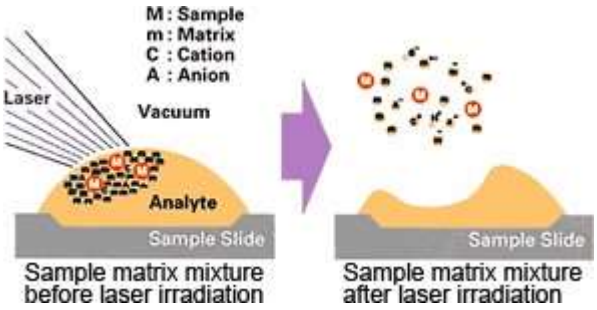
Ions that have successfully been deflected by the mass analyzers then hit the ion detector. Most often, these detectors are electron multipliers or microchannel plates, which emit a cascade of electrons when each ion hits the detector plate. This cascade results in amplification of each ion hit for improved sensitivity. This entire process is performed under an extreme vacuum (10^{-6} to 10^{-8} torr) to remove gas molecules, neutrals, and contaminating non-sample ions, which can collide with sample ions and alter their paths or produce non-specific reaction products.

Newer orbitrap technology captures ions around a central spindle electrode and then analyzes their m/z values as they move across the spindle with different harmonic oscillation frequencies. Orbitrap technology can achieve extremely high sensitivity and resolution of obtained mass spectra.

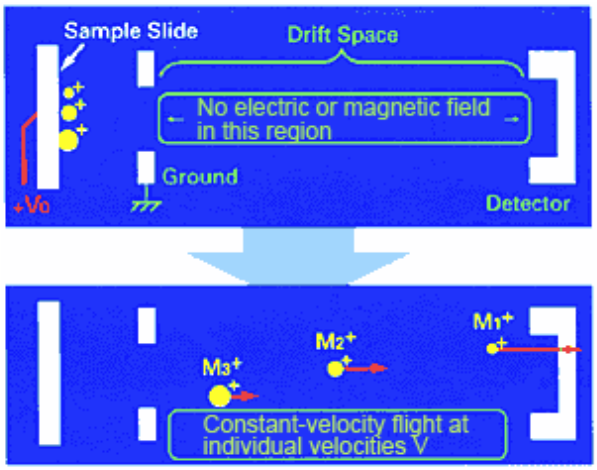
Mass spectrometers are connected to computers with software that analyzes the ion detector data and produces graphs that organize the detected ions by their individual *m/z* and relative abundance. These ions can then be processed through databases to predict the identity of the molecule based on the *m/z*.

MALDI-TOF

MALDI is the abbreviation for "Matrix Assisted Laser Desorption/Ionization." The sample for MALDI is uniformly mixed in a large quantity of matrix. The matrix absorbs the ultraviolet light (nitrogen laser light, wavelength 337 nm) and converts it to heat energy. A small part of the matrix (down to 100 nm from the top outer surface of the Analyte in the diagram) heats rapidly (in several nano seconds) and is vaporized, together with the sample.



TOF MS is the abbreviation for Time of Flight Mass Spectrometry. Charged ions of various sizes are generated on the sample slide, as shown in the diagram. A potential difference V_0 between the sample slide and ground attracts the ions in the direction shown in the diagram. The velocity of the attracted ions v is determined by the law of conservation of energy. As the potential difference V_0 is constant with respect to all ions, ions with smaller m/z value (lighter ions) and more highly charged ions move faster through the drift space until they reach the detector. Consequently, the time of ion flight differs according to the mass-to-charge ratio (m/z) value of the ion. The method of mass spectrometry that exploits this phenomenon is called Time of Flight Mass Spectrometry.



METHODS TO DETERMINE SUPER SECONDARY STRUCTURES

The methods used for determining the secondary structures are

1. statistical methods
2. Pattern based methods
3. Statistical-mechanical methods

Used to identify secondary structure (alpha, beta and coil) elements from protein sequence.

Chou-fasman method:

The Chou–Fasman method is an empirical technique for the prediction of secondary structures in proteins, originally developed in the 1970s by Peter Y. Chou and Gerald D. Fasman. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters were derived for the appearance of each amino acid in each secondary structure type, and these parameters are used to predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein. The method is at most about 50–60% accurate in identifying correct secondary structures,which is significantly less accurate than the modern machine learning–based techniques.

The original Chou–Fasman parameters found some strong tendencies among individual amino acids to prefer one type of secondary structure over others. Alanine, glutamate, leucine, and methionine were identified as helix formers, while proline and glycine, due to the unique conformational properties of their peptide bonds, commonly end a helix. The original Chou–Fasman parameters were derived from a very small and non-representative sample of protein structures due to the small number of such structures that were known at the time of their original work. These original parameters have since been shown to be unreliable and have been updated from a current dataset, along with modifications to the initial algorithm.

The Chou–Fasman method takes into account only the probability that each individual amino acid will appear in a helix, strand, or turn. Unlike the more complex GOR method, it does not reflect the conditional probabilities of an amino acid to form a particular secondary structure given that its neighbors already possess that structure. This lack of cooperativity increases its computational efficiency but decreases its accuracy, since the propensities of individual amino acids are often not strong enough to render a definitive prediction.

The Chou–Fasman method predicts helices and strands in a similar fashion, first searching linearly through the sequence for a "nucleation" region of high helix or strand probability and then extending the region until a subsequent four-residue window carries a probability of less than 1. As originally described, four out of any six contiguous amino acids were sufficient to nucleate helix, and three out

of any contiguous five were sufficient for a sheet. The probability thresholds for helix and strand nucleations are constant but not necessarily equal; originally 1.03 was set as the helix cutoff and 1.00 for the strand cutoff.

Turns are also evaluated in four-residue windows, but are calculated using a multi-step procedure because many turn regions contain amino acids that could also appear in helix or sheet regions. Four-residue turns also have their own characteristic amino acids; proline and glycine are both common in turns. A turn is predicted only if the turn probability is greater than the helix or sheet probabilities *and* a probability value based on the positions of particular amino acids in the turn exceeds a predetermined threshold. The turn probability p(t) is determined as:

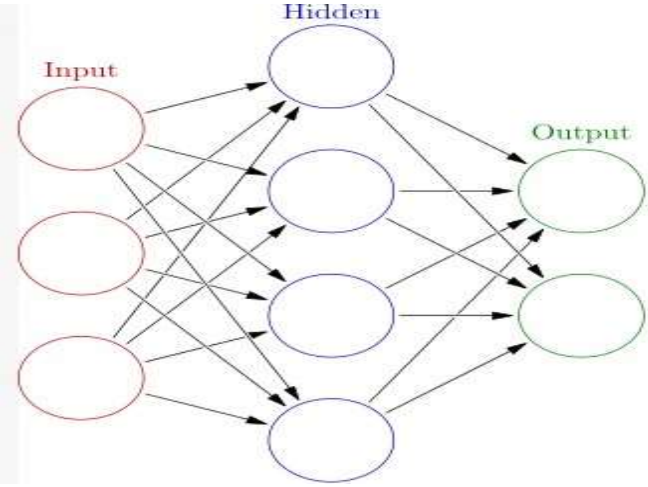
$$p(t) = p_t(j) \times p_t(j+1) \times p_t(j+2) \times p_t(j+3)$$

where *j* is the position of the amino acid in the four-residue window. If p(t) exceeds an arbitrary cutoff value (originally 7.5e–3), the mean of the p(j)'s exceeds 1, and p(t) exceeds the alpha helix and beta sheet probabilities for that window, then a turn is predicted. If the first two conditions are met but the probability of a beta sheet p(b) exceeds p(t), then a sheet is predicted instead.

Artificial neural networks:

Artificial neural networks are a class of general computational structure based based on the anatomy and physiology of biological nervous system. They have been applied successfully to a wide variety of pattern recognition, classification and decision problems. The general architecture of a neural network model consists of three layers of processing units or nodes. This machine learning algorithm is mainly used for secondary structure prediction.

i) Input layer
ii) Hidden layer
iii) Output layer



SUPERSECONDARY STRUCTURE

Introduction

Secondary structure elements are observed to combine in specific geometric arrrangements known as motifs or supersecondary structures. In this section we will look at motifs consisting of no more than three secondary structure elements. Larger motifs such as the greek key will be examined in the sections on tertiary structure and protein folds.

Beta-hairpins

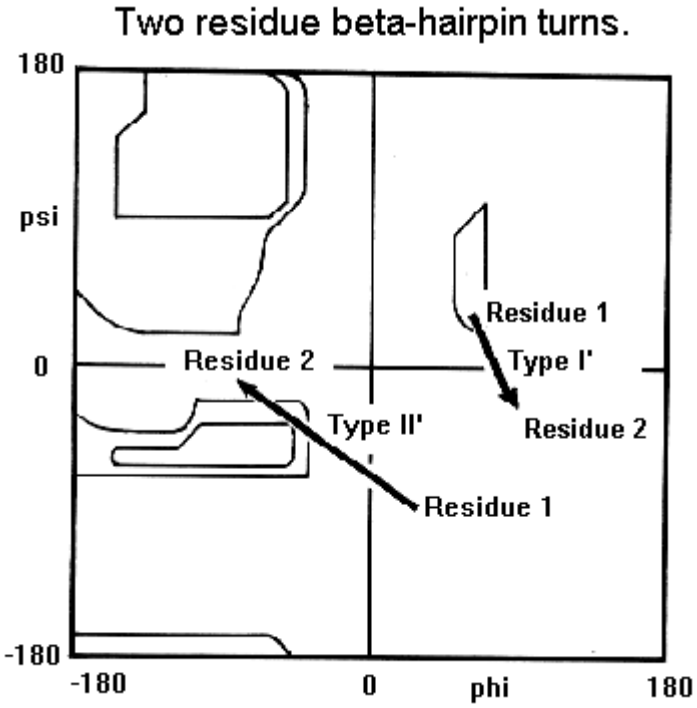
Beta-hairpins are one of the simplest supersecondary structures and are widespread in globular proteins. They occur as the short loop regions between antiparallel hydrogen bonded beta-strands. In general a reverse turn (or beta-turn, as it they are sometimes called) is any region of a protein where there is a hydrogen bond involving the carbonyl of residue i and the NH group of residue i+3. An alternative definition states that the alpha-carbons of residues i and i+3 must be within 7.0 Angstroms. The structures of reverse turns are outlined in the section on peptide geometry. In this section we will concentrate on those turns which occur between consecutive beta-strands, known as beta-hairpins. Sibanda and Thornton have devised a system for classifying beta-hairpins which is based on two conventions for defining loop regions. In this section we will not go into such details as the objective is indicate the most commonly observed hairpin loop structures.

Beta-hairpin loops adopt specific conformations which depend on their lengths and sequences. Sibanda and Thornton have shown that 70% of beta-hairpins are less than 7 residues in length with the two-residue turns forming the most noticeable component. These two-residue beta-hairpins all adopt one of the classical reverse turn conformations with an obvious preference for types I' and II'. Type I 2-residue hairpins also occur but with lower abundance. This contrasts with reverse turns where types I and II tend to dominate. In beta-hairpins the type I' turn has the correct twist to match the twist of the beta-sheet and modelling studies indicated that if either type I or type II turns were to connect the antiparallel beta-strands, they would diverge within a short distance from the turn.

Two-residue beta-hairpins

1. **Type I'**.The first residue in this turn adopts the left-handed alpha-helical conformation and therefore shows preference for glycine, asparagine or aspartate. These residues can adopt conformations with positive phi angles due to the absence of a side chain with glycine and because of hydrogen bonds between the side chain and main chain in the case of asparagine or aspartate. The second residue of a type I' turn is nearly always glycine as the required phi and psi angles are well outside the allowed regions of the Ramachandran plot for amino acids with side chains. Were another type of amino acid to occur here there would be steric hindrance between its side chain and the carbonyl oxygen of the preceding residue..

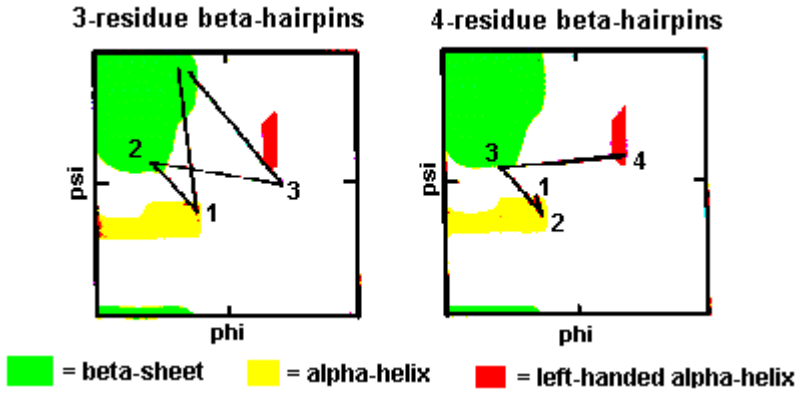
2. **Type II'**. The first residue of these turns has a conformation which can only be adopted by glycine (see below Ramachandran plot). The second residue shows a preference for polar amino acids such as serine and threonine.



3. **Type I**. Both residues of these turns adopt alpha-helical conformations.

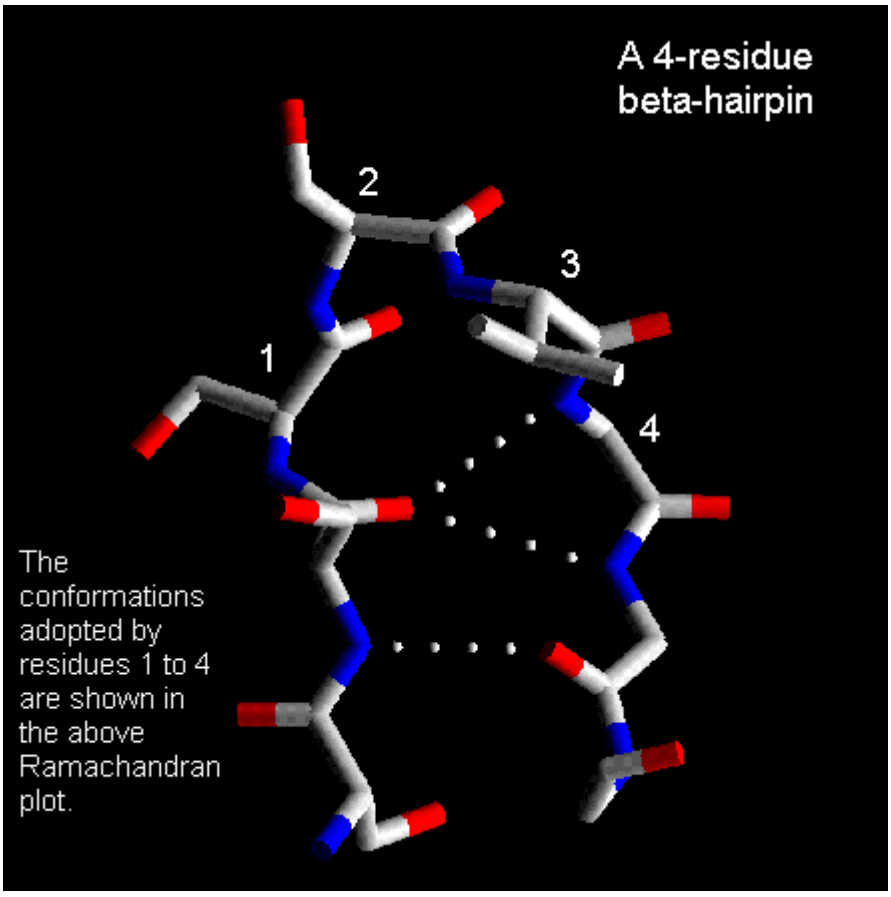
Three-residue beta-hairpins

Normally the residues at the ends of the two beta-strands only make one hydrogen bond as shown below. The intervening three residues have distinct conformational preferences as shown in the Ramachandran plot. The first residue adopts the right-handed alpha-helical conformation and the second amino acid lies in the bridging region between between alpha-helix and beta-sheet. Glycine, asparagine or aspartate are frequently found at the last residue position as this adopts phi and psi angles close to the left-handed helical conformation. For RasMol-MIME.



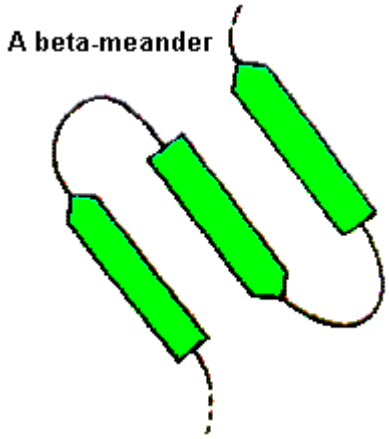
Four-residue beta-hairpins

These are also quite common with the first two residues adopting the alpha-helical conformation. The third residue has phi and psi angles which lie in the bridging region between between alpha-helix and beta-sheet and the final residue adopts the left-handed alpha-helical conformation and is therefore usually Gly, Asp or Asn.



Longer loops

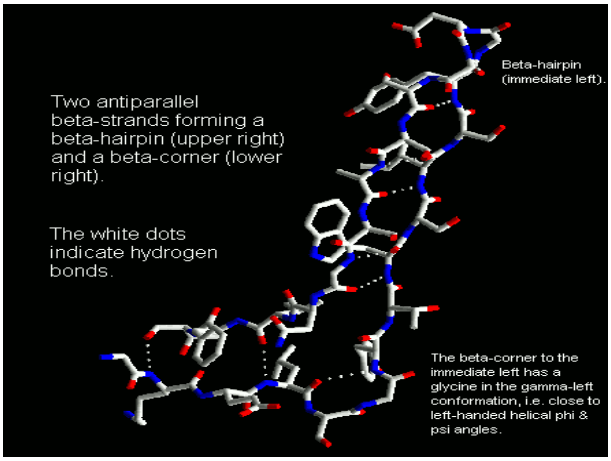
For these, a wide range of conformations is observed and the general term 'random coil' is sometimes used. Consecutive antiparallel beta-strands when linked by hairpins form a supersecondary structure known as the beta-meander.



Beta-corners

Beta-strands have a slight right-handed twist such that when they pack side-by-side to form a beta-sheet, the sheet has an overall left-handed curvature. Antiparallel beta-strands forming a beta-hairpin can accommodate a 90 degree change in direction known as a beta-corner. The strand on the inside of the bend often has a glycine at this position while the other strand can have a beta-bulge. The latter involves a single residue in the right-handed alpha-helical conformation which breaks the hydrogen bonding pattern of the beta-sheet. This residue can also be in the left-handed helical or bridging regions of the Ramachandran plot.

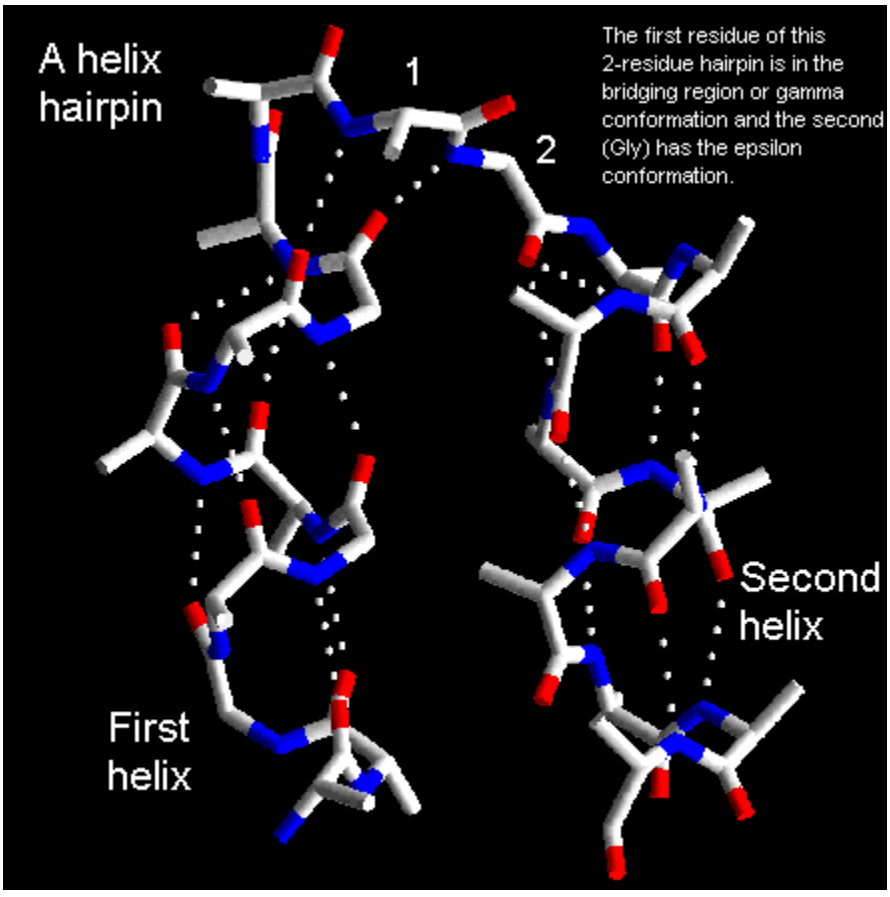
Beta-corners are observed to have a right-handed twist when viewed from the concave side.



Helix hairpins

A helix hairpin or alphaalpha-hairpin refers to the loop connecting two antiparallel alpha-helical segments. Clearly, the longer the length of the loop the greater the number of possible conformations. However, for short connections there are a limited number of conformations and for the shortest loops of two or three residues, there is only one allowed conformation. Antiparallel alpha-helices will interact generally by hydrophobic interactions between side chains at the interface. Therefore, hydrophobic amino acids have to be appropriately positioned in the amino acid sequence (one per turn of each helix) to generate a hydrophobic core. Efimov has analysed the conformations of alpha-alpha-hairpins and some of his results are summarised below.

The shortest alpha-helical connections involve two residues which are oriented approximately perpendicular to the axes of the helices. Analysis of known structures reveals that the first of these two residues adopts phi and psi angles in the bridging or alpha-helical regions of the Ramachandran plot. The second residue is always glycine and is in a region of the Ramachandran plot with positive phi which is not available to other amino acids.

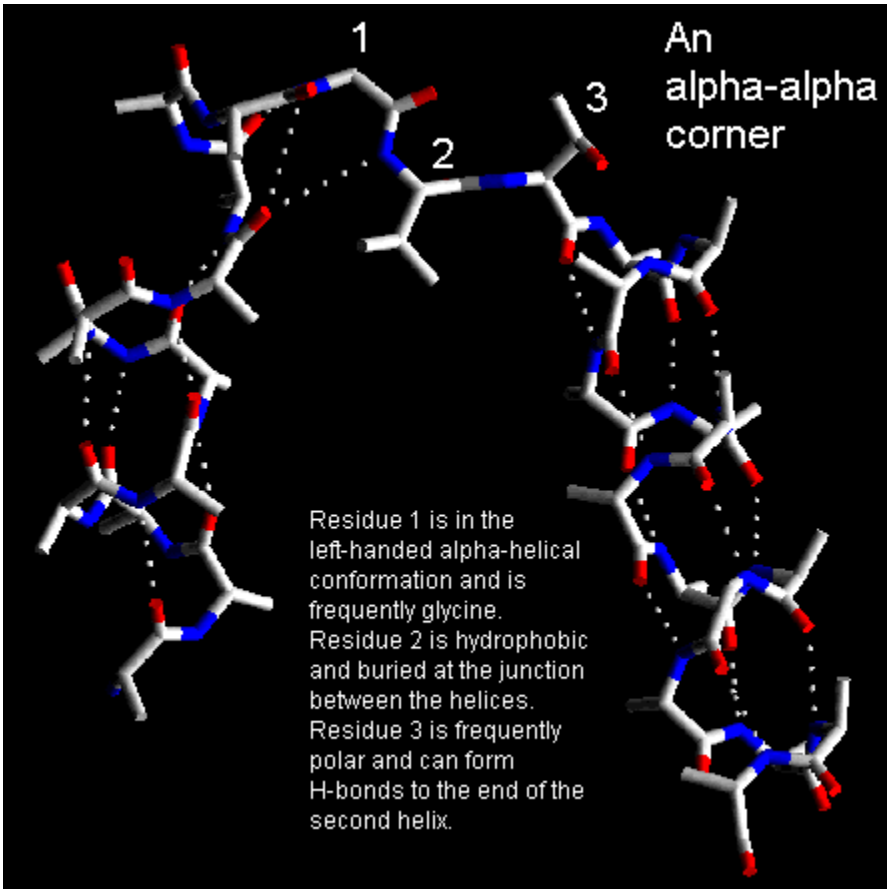


Three residue loops are also observed to have conformational preferences. The first residue occupies the bridging region of the Ramachandran plot, the second adopts the left-handed helical conformation and the last residue is in a beta-strand conformation.

Four-residue loops adopt one of two possible conformations. One is similar to the three residue loop conformation described above except that there is an additional residue in the beta-strand conformation at the fourth position. The other conformation involves the four residues adopting bridging, beta, bridging, and beta conformations, respectively.

Alpha-alpha corner

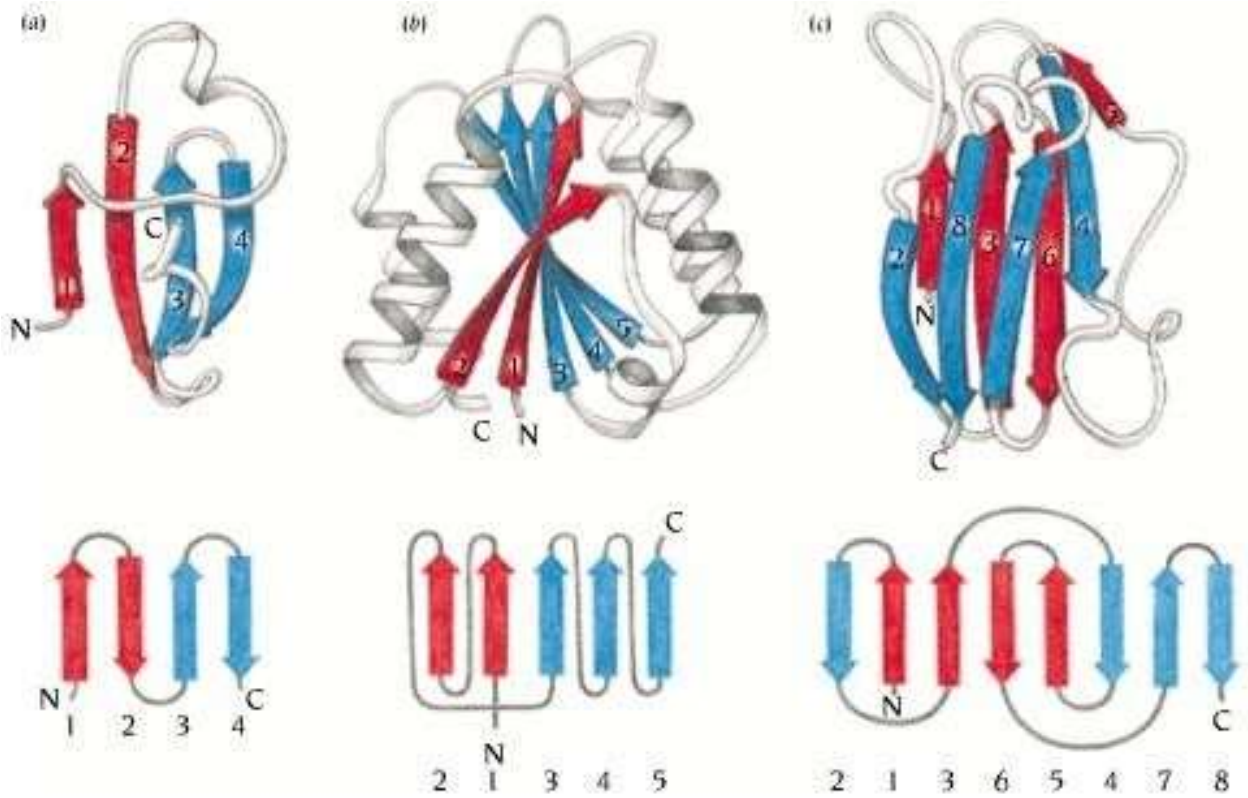
Short loop regions connecting helices which are roughly perpendicular to one another are refered to as alpha-alpha-corners. Efimov has shown that the shortest alpha-alpha-corner has its first residue in the left-handed alpha-helical conformation and the next two residues in beta-strand conformations. This conformation can only be adopted when the two helices form a right-handed corner. Indeed, if the helices were linked to form a left-handed corner there would be steric hindrance. This may explain the scarcity of left-handed alpha-alpha corners in protein X-ray structures.



The C-terminal residue of the first helix, which is in the left-handed alpha-helical conformation, must have a short side chain to avoid steric hindrance and is observed commonly to be glycine. The first residue of the second helix, which is in the beta-conformation, frequently has a small polar side chain such as Ser or Asp which can form hydrogen bonds with the free NH groups at the amino-terminal end of the second helix. The central residue of the alpha-alpha-corner is almost always hydrophobic as it is buried and interacts with other non-polar side chains buried where the ends of the two helices contact each other.

Topology diagrams, up and down & TIM barrel structures nucleotide binding folds.

Describes how secondary structure elements in a particular protein are linked, and the order of β strands in a β sheet structure. Schematic representation of protein topology Topology diagrams help illustrate how the 3D structure of a long polypeptide chain is constructed from its secondary structure elements.



- Beta sheets are usually represented simply by arrows in topology diagrams that show both the direction of each β strand and the way the strands are connected to each other along the polypeptide chain.

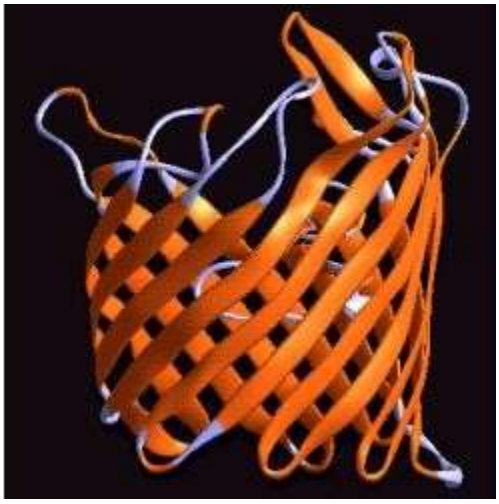
Barrel structures

In molecular biology the TIM barrel is a conserved protein fold consisting of eight α-helices and eight parallel β-strands that alternate along the peptide backbone. The structure is named after triosephosphate

isomerase, a conserved metabolic enzyme. TIM barrels are one of the most common protein folds. One of the most intriguing features among members of this class of proteins is although they all exhibit the same tertiary fold there is very little sequence similarity between them. At least 15 distinct enzyme families use this framework to generate the appropriate active site geometry, always at the C-terminal end of the eight parallel beta-strands of the barrel.

TIM barrels are considered α/β protein folds because they include an alternating pattern of α-helices and β-strands in a single domain. In a TIM barrel the helices and strands (usually 8 of each) form a solenoid that curves around to close on itself in a doughnut shape, topologically known as a toroid. The parallel β-strands form the inner wall of the doughnut (hence, a β-barrel), whereas the α-helices form the outer wall of the doughnut. Each β-strand connects to the next adjacent strand in the barrel through a long right-handed loop that includes one of the helices, so that the ribbon N-to-C coloring in the top view proceeds in rainbow order around the barrel. The TIM barrel can also be thought of, then, as made up of 8 overlapping, right-handed β-α-β super-secondary structures .

Although the ribbon diagram shows a hole in the protein's central core, the amino acid side chains are not shown in this representation. The protein's core is actually tightly packed, mostly with bulky hydrophobic amino acid residues although a few glycines are needed to allow wiggle room for the highly constrained center of the 8 approximate repeats to fit together. The packing interactions between the strands and helices are also dominated by hydrophobicity and the branched aliphatic residues valine, leucine, and isoleucine comprise about 40% of the total residues in the β-strands.



The up-and-down beta-barrel is a common folding motif found frequently in proteins that bind and transport hydrophobic ligands. It is formed by an array of beta-strands arranged in an antiparallel manner with each strand hydrogen-bonded to neighboring strands nearly always adjacent in the amino acid sequence. The arrangement is completed by forming hydrogen bonds between the first and last strands. The barrel motif so formed produces interior and exterior components. Proteins belonging to this class of up-and-down beta-

barrels are found typically to be lipid-binding proteins in which the interior surface forms a cavity or pit that serves as the ligand binding region. Two evolutionarily distinct but structurally related families of such carriers have been identified by comparing known crystal structures. One group found intracellularly uses a 10-stranded beta-structure and a second family of proteins typically found extracellularly utilizes an 8-stranded motif. The 10-stranded beta-barrels have a large, hydrophilic water-filled interior cavity that serves as the ligand-binding domain. Hydrophobic lipids such as fatty acids and retinoids bind within the cavity, totally sequestered from the external milieu. The 8-stranded beta-barrel proteins have a hydrophobic pit, which serves as the ligand-binding domain for compounds such as bilins and retinoids. The up-and-down beta-barrel motif appears to be one of nature's primary choices for hydrophobic ligand transport proteins.

TERTIARY STRUCTURE

The term ‘‘*tertiary structure*’’ refers to the arrangement of secondary structure elements and amino acid side chain interactions that define the three-dimensional structure of the folded protein. Imagine that a newly synthesized protein exists in nature as a fully extended polypeptide chain—it is said then to be *unfolded.* Now suppose that this protein is placed under the set of conditions that will lead to the formation of elements of secondary structure at appropriate locations along the polypeptide chain. Next, the individual elements of secondary structure arrange themselves in three-dimensional space, so that specific contacts are made between amino acid side chains and between backbone groups. The resulting *folded* structure of the protein is referred to as its *tertiary structure.*

Tertiary protein structure refers to the complete three dimensional folding of a protein. Stabilization of a protein's tertiary structure may involve interactions between amino acids located far apart along the primary sequence. These may include:

Weak interactions such as hydrogen bonds and Van der Waals interactions.

Ionic bonds involving negatively charged and positively charged amino acid side-chain groups.

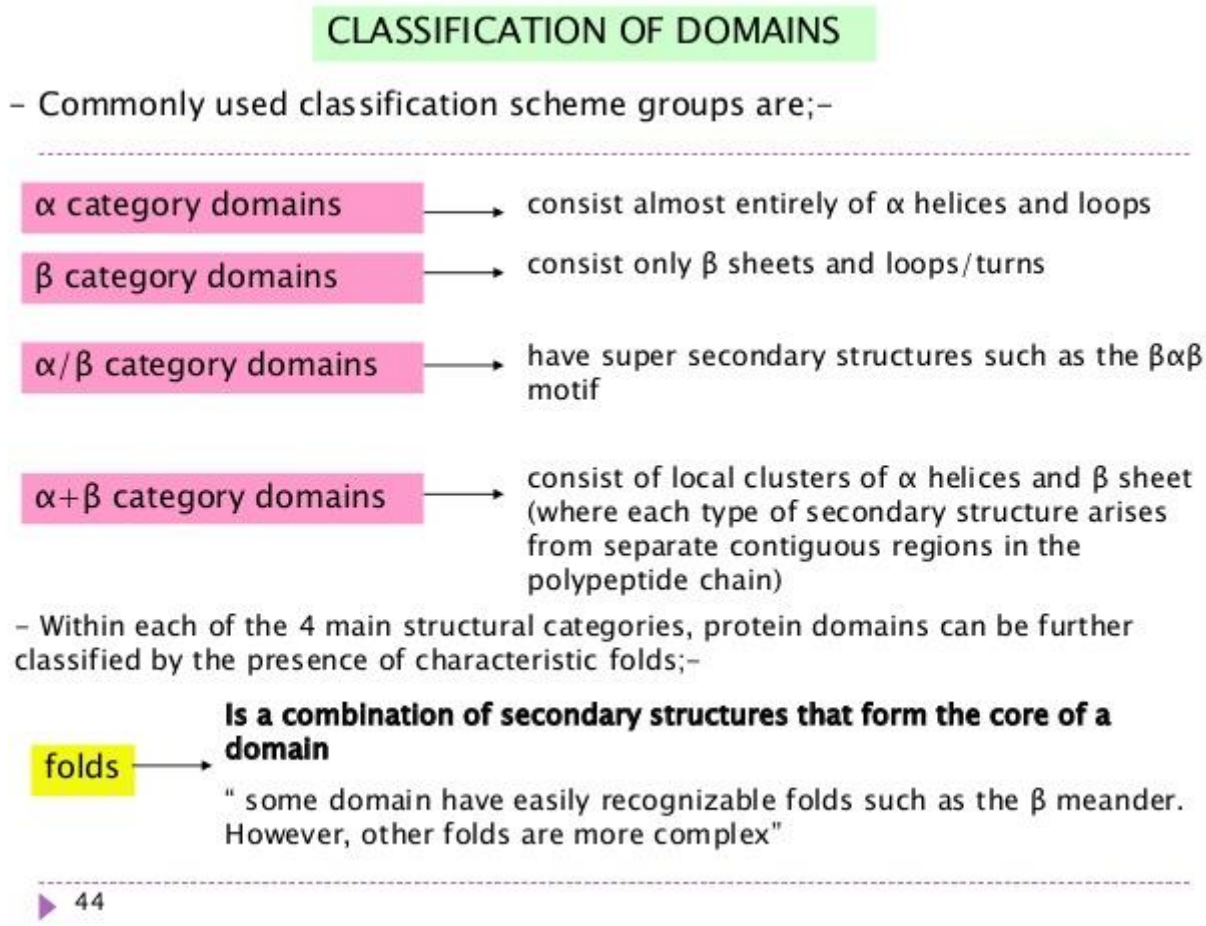
disulfide bonds, covalent linkages that may form as the thiol groups of two cysteine residues are oxidized to a disulfide:

2 R-SH R-S-S-R.

Interaction with the aqueous solvent known as the hydrophobic effect results in residues with non-polar side-chains typically buried in to the interior of a protein. Conversely, polar amino acid side-chains tend to on the surface of a protein where they are exposed to the aqueous milieu. However, many exceptions in which polar residues are buried or non-polar residues exposed on the surface of a protein. Such a typical locations might be stabilized, e.g., by interaction of amino acid side-chains with enzyme prosthetic groups or other ligands, by interactions between amino acid side-chains, or by association of proteins with lipid membranes, etc.

Domains

In some proteins one finds discrete regions of compact tertiary structure that are separated by stretches of the polypeptide chain in a more flexible arrangement. These discrete folded units are known as *domains*, and often they define functional units of the protein. For example, many cell membrane receptors play a role in signal transduction by binding extra cellular ligands at the cell surface. In response to ligand binding, the receptor undergoes a structural change that results in macromolecular interactions between the receptor and other proteins within the cell cytosol. These interactions in turn set off a cascade of biochemical events that ultimately lead to some form of cellular response to ligand binding. To function in this capacity, such a receptor requires a minimum of three separate domains: an extra cellular ligand binding domain, a transmembrane domain that anchors the protein within the cell membrane, and an intracellular domain that forms the locus for protein—protein interactions. Many enzymes are composed of discrete domains as well. For example, the crystal structure of the integral membrane enzyme prostaglandin synthase structure reveals three separate domains of the folded enzyme monomer: a -sheet domain that functions as an interface for dimerization with another molecule of the enzyme, a membrane-incorporated helical domain that anchors the enzyme to the biological membrane, and a extra membrane globular (i.e., compact folded region) domain that contains the enzymatic active site and is thus the catalytic unit of the enzyme.



Protein folding pathways

Protein folding is the physical process by which a polypeptide folds into its characteristic three-dimensional structure. Each protein begins as a polypeptide, translated from a sequence of mRNA as a linear chain of amino acids. This polypeptide lacks any developed three-dimensional structure (the left hand side of the neighboring figure). However each amino acid in the chain can be thought of having certain 'gross' chemical features. These may be hydrophobic, hydrophilic, or electrically charged, for example. These interact with each other and their surroundings in the cell to produce a well-defined, three dimensional shape, the folded protein (the right hand side of the figure), known as the native state. The resulting three-dimensional structure is determined by the sequence of the amino acids. The mechanism of protein folding is not completely understood.

Experimentally determining the three dimensional structure of a protein is often very difficult and expensive. However the sequence of that protein is often known. Therefore scientists have tried to use different biophysical techniques to manually fold a protein. That is, to predict the structure of the complete protein from the sequence of the protein.For many proteins the correct three dimensional structure is essential to function. Failure to fold into the intended shape usually produces inactive proteins with different properties. Several neurodegenerative and other diseases are believed to result from the accumulation of *misfolded* (incorrectly folded) proteins.

The relationship between folding and amino acid sequence

The amino-acid sequence (or primary structure) of a protein predisposes it towards its native conformation or conformations. It will fold spontaneously during or after synthesis. While these macromolecules may be regarded as "folding themselves", the mechanism depends equally on the characteristics of the cytosol,

including the nature of the primary solvent (water or lipid), the concentration of salts, the temperature, and molecular chaperones. Most folded proteins have a hydrophobic core in which side chain packing stabilizes the folded state, and charged or polar side chains on the solvent-exposed surface where they interact with surrounding water molecules. It is generally accepted that minimizing the number of hydrophobic sidechains exposed to water is the principal driving force behind the folding process, although a recent theory has been proposed which reassesses the contributions made by hydrogen bonding.

The process of folding *in vivo* often begins co-translationally, so that the N-terminus of the protein begins to fold while the C-terminal portion of the protein is still being synthesized by the ribosome. Specialized proteins called chaperones assist in the folding of other proteins. A well studied example is the bacterial GroEL system, which assists in the folding of globular proteins. In eukaryotic organisms chaperones are known as heat shock proteins. Although most globular proteins are able to assume their native state unassisted, chaperone-assisted folding is often necessary in the crowded intracellular environment to prevent aggregation; chaperones are also used to prevent misfolding and aggregation which may occur as a consequence of exposure to heat or other changes in the cellular environment.

For the most part, scientists have been able to study many identical molecules folding together *en masse*. At the coarsest level, it appears that in transitioning to the native state, a given amino acid sequence takes on roughly the same route and proceeds through roughly the same intermediates and transition states. Often folding involves first the establishment of regular secondary and supersecondary structures, particularly alpha helices and beta sheets, and afterwards tertiary structure. Formation of quaternary structure usually involves the "assembly" or "coassembly" of subunits that have already folded. The regular alpha helix and beta sheet structures fold rapidly because they are stabilized by intramolecular hydrogen bonds, as was first characterized by Linus Pauling. Protein folding may involve covalent bonding in the form of disulfide bridges formed between two cysteine residues or the formation of metal clusters. Shortly before settling into their more energetically favourable native conformation, molecules may pass through an intermediate "molten globule" state.

The essential fact of folding, however, remains that the amino acid sequence of each protein contains the information that specifies both the native structure and the pathway to attain that state: Folding is a spontaneous process independent of energy inputs from nucleoside triphosphates. The passage of the folded state is mainly guided by hydrophobic interactions, formation of intramolecular hydrogen bonds, and van der Waals forces, and it is opposed by conformational entropy, which must be overcome by extrinsic factors such as chaperones.

Disruption of the native state

In certain solutions and under some conditions proteins will not fold into their biochemically functional forms. Temperatures above (and sometimes those below) the range that cells tend to live in will cause proteins to unfold or "denature" (this is why boiling makes the white of an egg opaque). High concentrations of solutes, extremes of pH, mechanical forces, and the presence of chemical denaturants can do the same. A fully denatured protein lacks both tertiary and secondary structure, and exists as a so-called random coil. Under certain conditions some proteins can refold; however, in many cases denaturation is irreversible. Cells sometimes protect their proteins against the denaturing influence of heat with enzymes known as chaperones or heat shock proteins, which assist other proteins both in folding and in remaining folded. Some proteins never fold in cells at all except with the assistance of chaperone molecules, which either isolate individual proteins so that their folding is not interrupted by interactions with other proteins or help to unfold misfolded proteins, giving them a second chance to refold properly. This function is crucial to prevent the risk of precipitation into insoluble amorphous aggregates.

Incorrect protein folding and neurodegenerative disease

Misfolded proteins are responsible for prion-related illnesses such as Creutzfeldt-Jakob disease, bovine spongiform encephalopathy (mad cow disease), amyloid-related illnesses such as Alzheimer's Disease, and a number of other forms of proteopathy such as cystic fibrosis. These diseases are associated with the multimerization of misfolded proteins into insoluble, extracellular aggregates and/or intracellular inclusions; it is not clear whether the plaques are the cause or merely a symptom of illness.

Kinetics and the Levinthal Paradox

The entire duration of the folding process varies dramatically depending on the protein of interest. The slowest folding proteins require many minutes or hours to fold, primarily due to steric hindrances, and must pass through a number of intermediate states, like checkpoints, before the process is complete. On the other hand, very small single-domain proteins with lengths of up to a hundred amino acids typically fold in a single step. Time scales of milliseconds are the norm and the very fastest known protein folding reactions are complete within a few microseconds.

The Levinthal paradox observes that if a protein were to fold by sequentially sampling all possible conformations, it would take an astronomical amount of time to do so, even if the conformations were sampled at a rapid rate (on the nanosecond or picosecond scale). Based upon the observation that proteins fold much faster than this, Levinthal then proposed that a random conformational search does not occur in folding, and the protein must, therefore, fold by a directed process.

Techniques for studying protein folding

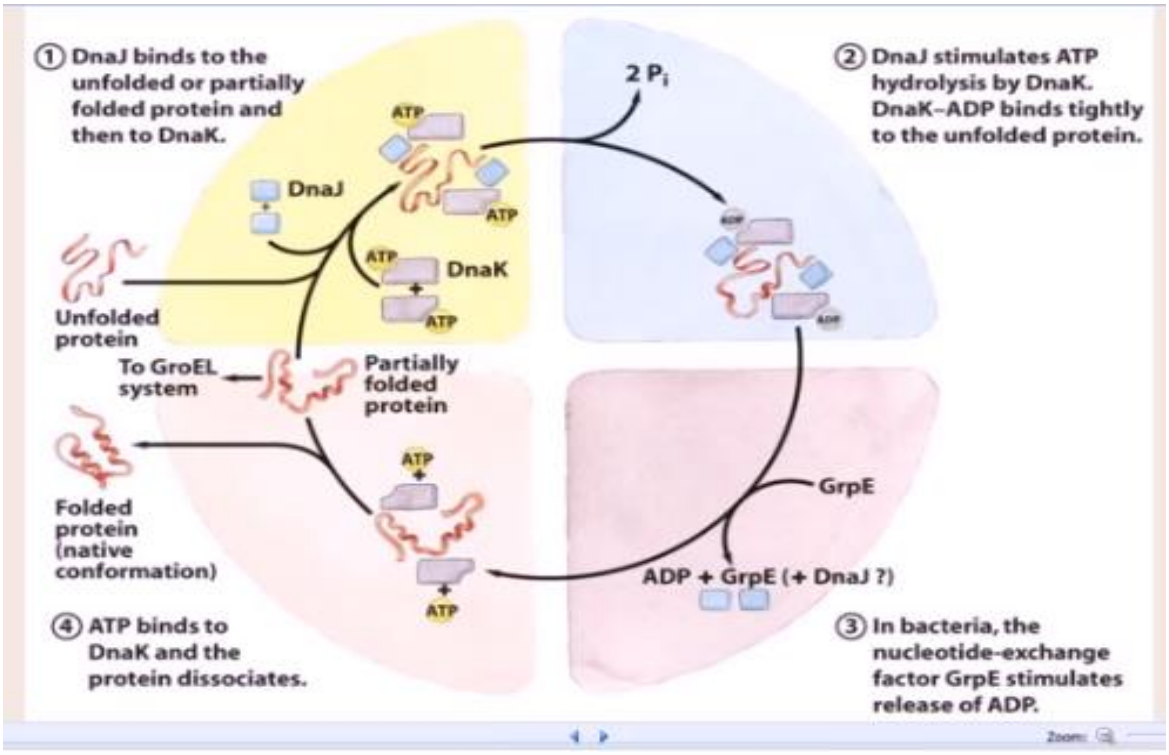
Modern studies of folding with high time resolution

The study of protein folding has been greatly advanced in recent years by the development of fast, time-resolved techniques. These are experimental methods for rapidly triggering the folding of a sample of unfolded protein, and then observing the resulting dynamics. Fast techniques in widespread use include ultrafast mixing of solutions, photochemical methods, and laser temperature jump spectroscopy.

Energy landscape theory of protein folding

The protein folding phenomenon was largely an experimental endeavor until the formulation of energy landscape theory by Joseph Bryngelson and Peter Wolynes in the late 1980s and early 1990s. This approach introduced the principle of minimal frustration, which asserts that evolution has selected the amino acid sequences of natural proteins so that interactions between side 19

chains largely favor the molecule's acquisition of the folded state. Interactions that do not favor folding are selected against, although some residual *frustration* is expected to exist. A consequence of these evolutionarily selected sequences is that proteins are generally thought to have globally "funneled energy landscapes" (coined by José Onuchic) that are largely directed towards the native state. This "folding funnel" landscape allows the protein to fold to the native state through any of a large number of pathways and intermediates, rather than being restricted to a single mechanism. The theory is supported by computational simulations of model proteins and has been used to improve methods for protein structure prediction and design.



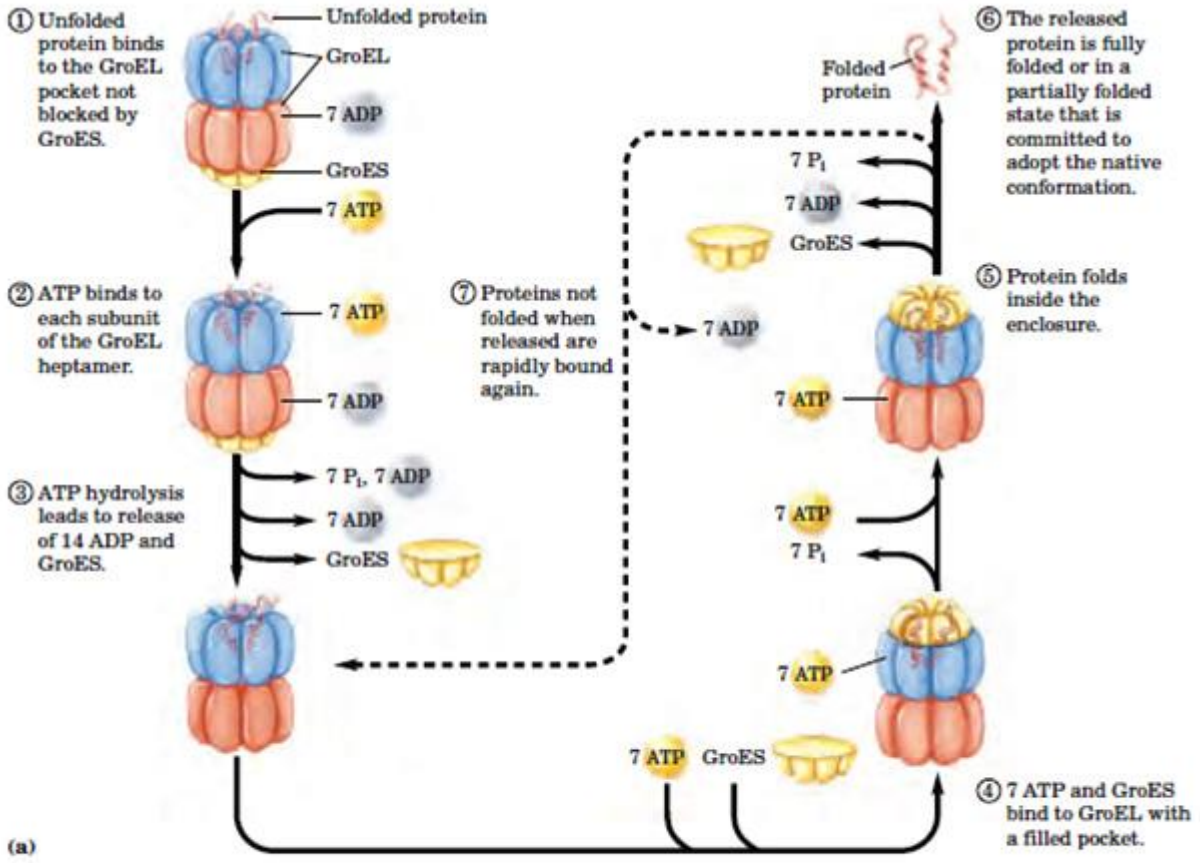
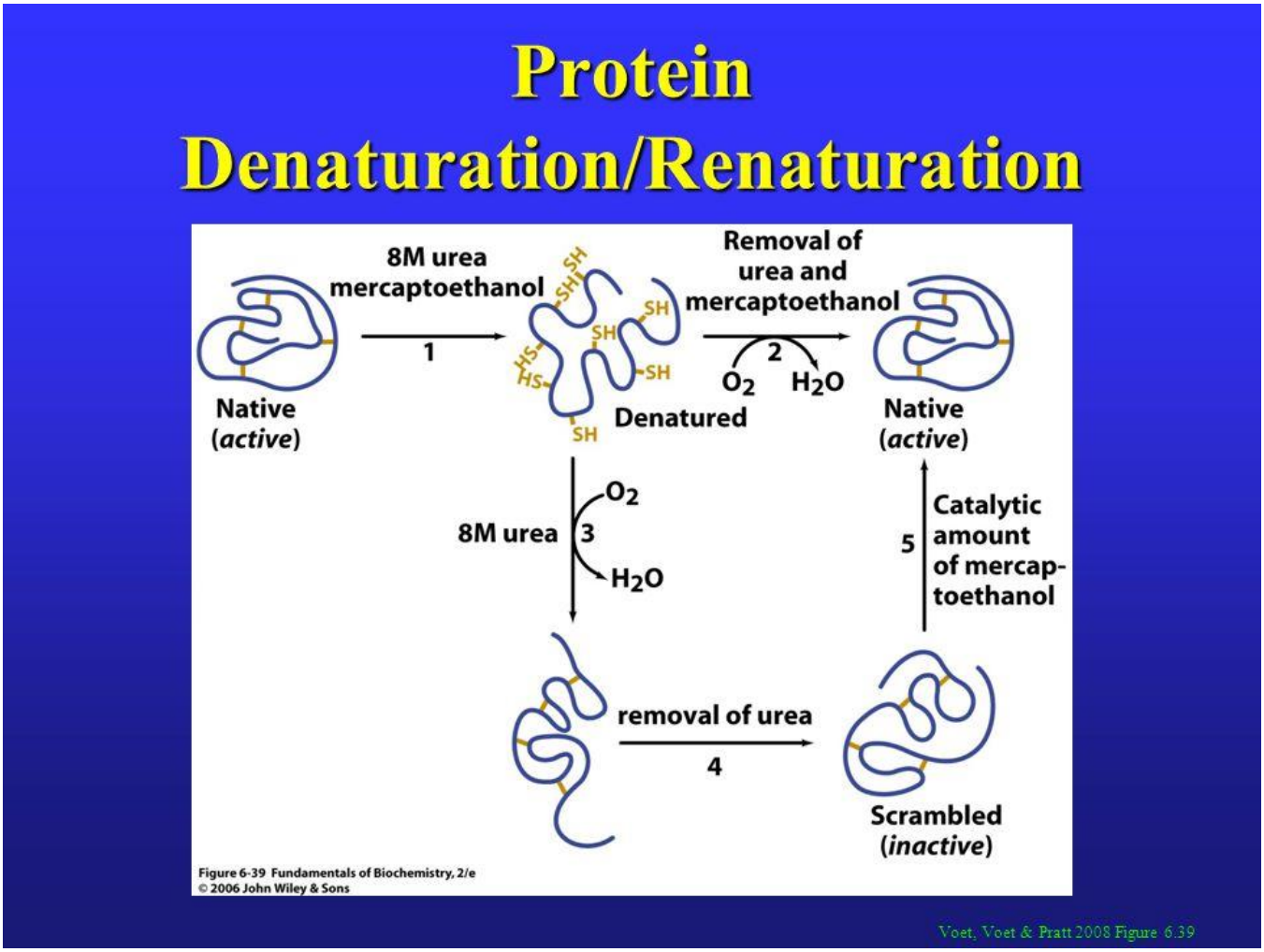


Figure 3.2 A proposed pathway for the action of the *E. coli* chaperonin GroEL (a member of the Hsp60 protein family) and GroES (Nelson and Cox, 2005).

DENATURATION

Proteins kept in solution for long periods may denature and lose their biological or enzymatic properties. Such loss of activity can be minimized when the solution is kept at low temperature. For example alkaline phosphatase in blood will be inactivated within two days when kept at 20oC; but will be active for 7 days at 4oC 20 days at-20oC and for three months at -80oC minimized. Brief heating, urea, salicylate, X-ray, ultraviolet rays, high pressure, vigorous shaking and such other physicochemical agents produce nonspecific alterations in secondary, tertiary and quaternary structures of protein molecules. This is called denaturation.

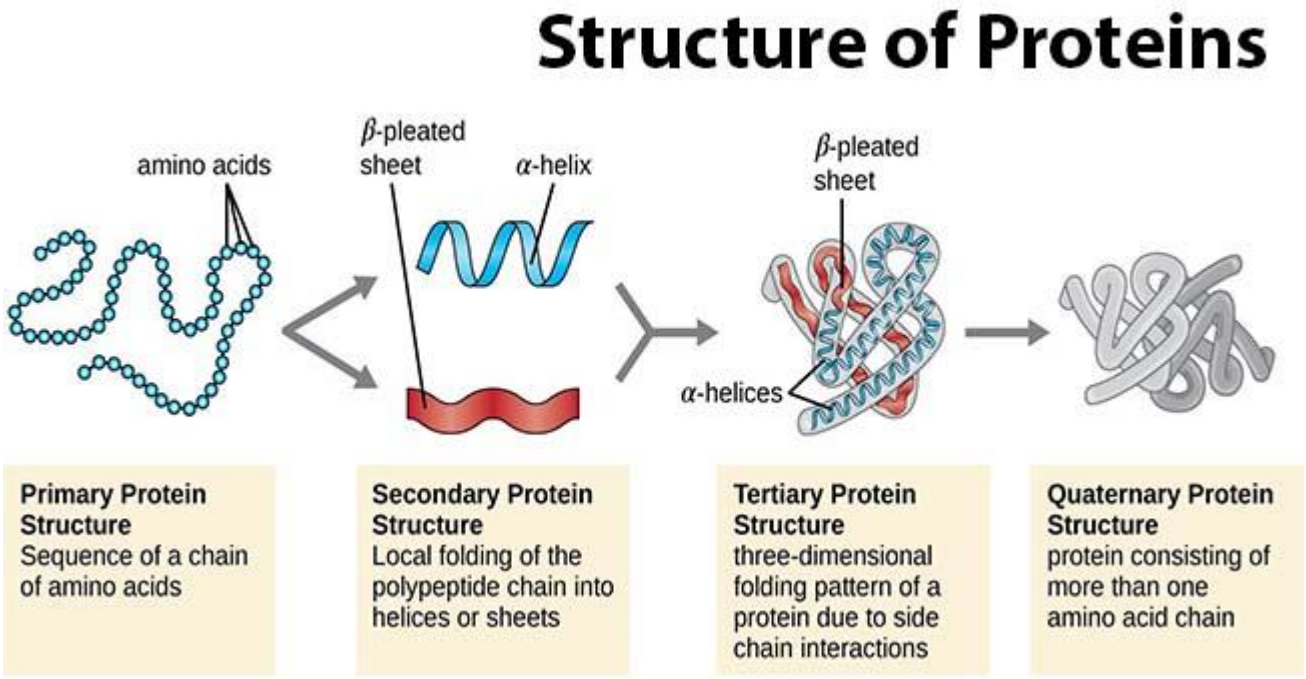


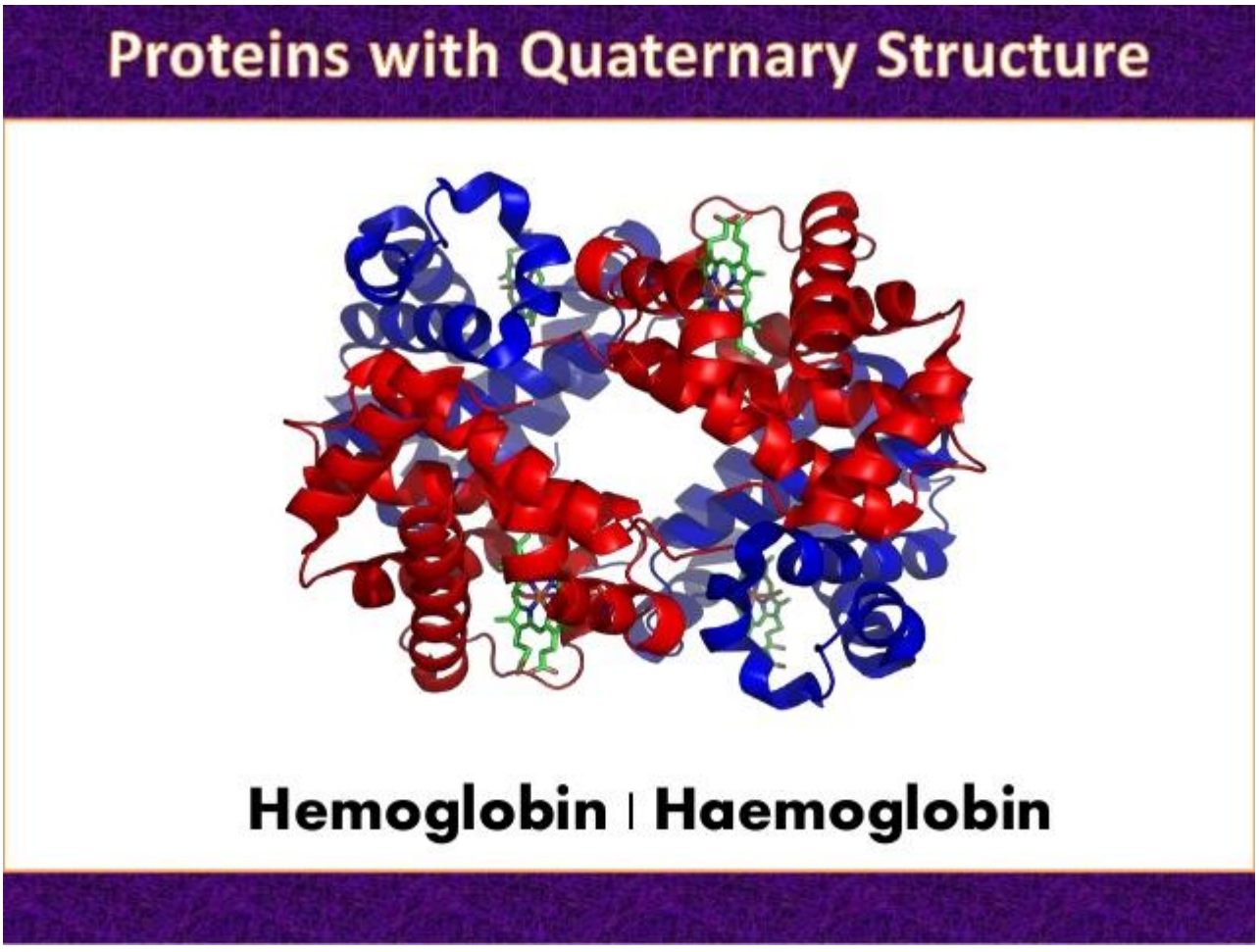
Primary structure is not altered during denaturation. It generally decreases the solubility, increases precipitability and often causes loss of biological activity. Native proteins are often resistant to proteolytic enzymes, but denatured proteins will have more exposed sites for enzyme action. Since cooking needs to denaturation of proteins, cooked foods are more easily digested. **Heat coagulation** When some proteins are heated at iso-electric point, denaturation takes place irreversibly to produce thick floating conglomerates called coagulum. This process is called heat coagulation. Albumin is easily coagulated, and globulin to a lesser extent. Some proteins when heated, though denatured, are still soluble, they may be precipitated by bringing to iso-electric pH. This is the basis of 'Heat and Acetic acid test', very commonly employed to detect the presence of albumin in urine. **Renaturation:** Denatured proteins are sometimes re-natured when the physical agent is removed. Ribonuclease is a good example for such reversible denaturation. Urea (8 Molar) is usually employed to dissociate polypeptide subunits of proteins. When urea is removed by dialysis, the subunits are re-associated and the biological activity is regained. Immunoglobulins show such reversible denaturation by urea. But many of the proteins (e. g) could not be denatured by removing the physical agent that denatured the protein

Proteins can be preserved for years when lyophilized. Lyophilisation (freeze drying) is the process by which water is evaporated at very low temperature in vacuum. Formaldehyde will fix the three dimensional structure of protein intact, and afterwards physicochemical agents will not denature the proteins. This is the basis of fixation of tissues by formalin before histopathological examination.

QUATERNARY STRUCTURE

Quaternary protein structure refers to the regular association of two or more polypeptide chains to form a complex. A multi-subunit protein may be composed of two or more identical polypeptides, or it may include different polypeptides. Quaternary structure tends to be stabilized mainly by weak interactions between residues exposed on surfaces polypeptides within a complex. A multimeric complex may be important to enzyme activity, e.g., in cases in which an active site is formed by residues from more than one polypeptide subunit, or when adjacent active sites may be involved sequentially in catalysis of a complex reaction. Regulation of functional activity may involve cooperative interactions among protein subunits in a complex. Many proteins exist in nature as oligomers, complexes composed of (often symmetric) noncovalent assemblies of two or more monomer subunits Most intracellular enzymes are oligomeric and may be composed either of a single type of monomer subunit (*homomultimers*) or of several different kinds of subunits (*heteromultimers*). The simplest case is a protein composed of identical subunits. (Eg) Liver alcohol dehydrogenase .The way in which separate folded monomeric protein subunits associate to form the oligomeric protein constitutes the quaternary structure of that protein.





Modular nature

Many proteins have a modular design with multiple distinct domains resulting from gene fusion events during evolution. A domain with a particular structure and function may be part of the structure of several otherwise distinct proteins. For example, an enzyme's primary structure may include a segment that folds to

produce an active site with particular catalytic activity, plus other segments that may mediate regulation of the enzyme or binding of the enzyme to a membrane. Nature's Modular Strategy for Protein Design Now that many thousands of proteins have been sequenced (more than 100,000 sequences are known), it has become obvious that certain protein sequences that give rise to distinct structural domains are used over and over again in modular fashion. These protein modules may occur in a wide variety of proteins, often being used for different purposes, or they may be used repeatedly in the same protein. the tertiary structures of five protein modules several proteins that contain versions of these modules. These modules typically contain about 40 to 100 amino acids and often adopt a stable tertiary structure when isolated from their parent protein. One of the best-known examples of a protein module is the immunoglobulin module, which has been found not only in immunoglobulins but also in a wide variety of cell surface proteins, including cell adhesion molecules and growth factor receptors, and even in *twitchin,* an intracellular protein found in muscle. It is likely that more protein modules will be identified. Some proteins play a recently discovered role in the complex pathways of cellular response to hormones and growth factors. These proteins, the scaffold or adapter proteins, have a modular organization in which specific parts (modules) of the protein's structure recognize and bind certain structural elements in other proteins through protein–protein interactions. For example, *SH2* modules bind to proteins in which a tyrosine residue has become phosphorylated on its phenolic OOH, and *SH3* modules bind to proteins having a characteristic grouping of proline residues. Others include *PH* modules, which bind to membranes, and *PDZ*-containing proteins, which bind specifically to the C-terminal amino acid of certain proteins. Because scaffold proteins typically possess several of these different kinds of modules, they can act as a scaffold onto which a set of different proteins is assembled into a multiprotein complex. Such assemblages are typically involved in coordinating and communicating the many intracellular responses to hormones or other signaling molecules.

Complex formation

Many proteins exist in nature as **oligomers,** complexes composed of (often symmetric) noncovalent assemblies of two or more monomer subunits. In fact, subunit association is a common feature of macromolecular organization in biology. Most intracellular enzymes are oligomeric and may be composed either of a single type of monomer subunit (*homomultimers*) or of several different kinds of subunits (*heteromultimers*). The simplest case is a protein composed of identical subunits. Liver alcohol dehydrogenase, is such a protein. More complicated proteins may have several different subunits in one, two, or more copies. Hemoglobin, for example, contains two each of two different subunits and is referred to as an 22-complex. An interesting counterpoint to these relatively simple cases is made by the proteins that form polymeric structures. Tubulin is an dimeric protein that polymerizes to form microtubules of the formula ( )*n*. The way in which separate folded monomeric protein subunits associate to form the oligomeric protein constitutes the quaternary structure of that protein.

The subunits of an oligomeric protein typically fold into apparently independent globular conformations and then interact with other subunits. The particular surfaces at which protein subunits interact are similar in nature to the interiors of the individual subunits. These interfaces are closely packed and involve both polar and hydrophobic interactions. Interacting surfaces must therefore possess complementary arrangements of polar and hydrophobic groups. Oligomeric associations of protein subunits can be divided into those between identical subunits and those between nonidentical subunits. Interactions among identical subunits can be further distinguished as either **isologous** or **heterologous.** In isologous interactions, the interacting surfaces are identical, and the resulting structure is necessarily dimeric and closed, with a twofold axis of symmetry. If any additional interactions occur to form a trimer or tetramer, these must use different

interfaces on the protein's surface. Many proteins, including concanavalin and prealbumin, form tetramers by means of two sets of isologous interactions. In contrast, heterologous associations among subunits involve nonidentical interfaces. These surfaces must be complementary, but they are generally not symmetric. heterologous interactions are necessarily open-ended. This can give rise either to a closed cyclic structure, if geometric constraints exist, or to large polymeric assemblies. The closed cyclic structures are far more common and include the trimers of aspartate transcarbamoylase catalytic subunits and the tetramers of neuraminidase and hemerythrin.

Overview of methods to determine 3D structures

Protein nuclear magnetic resonance spectroscopy (usually abbreviated **protein NMR**) is a field of structural biology in which NMR spectroscopy is used to obtain information about the structure and dynamics of proteins. The field was pioneered by, among others, Kurt Wüthrich, who shared the Nobel Prize in Chemistry in 2002. Protein NMR techniques are continually being used and improved in both academia and the biotech industry. Structure determination by NMR spectroscopy usually consists of several following phases, each using a separate set of highly specialized techniques. The sample is prepared, resonances are assigned, restraints are generated and a structure is calculated and validated.

Protein nuclear magnetic resonance is performed on aqueous samples of highly purified protein. Usually the sample consist of between 300 and 600 microlitres with a protein concentration in the range 0.1 – 3 millimolar. The source of the protein can be either natural or produced in an expression system using recombinant DNA techniques through genetic engineering. Recombinantly expressed proteins are usually easier to produce in sufficient quantity, and makes isotopic labelling possible.

The most abundant isotopes of carbon and oxygen, carbon-12 and oxygen-16, have no net nuclear spin, which is the physical property nuclear magnetic resonance spectroscopy exploits. The most abundant isotopes of nitrogen, nitrogen-14, has a net nuclear spin; however, it also has a large quadrupolar moment, a property of the atomic nuclei which prevents high-resolution information to be obtained from this isotope. Thus nuclear magnetic resonance of proteins from natural sources are restricted to utilizing nuclear magnetic resonance based solely on protons. However the less common isotopes, carbon-13 and nitrogen-15, are suitable for nuclear magnetic resonance, and therefore labelling the proteins with these compounds open up possibilities for doing more advanced experiments which also detect or use these nuclei. Isotopical labelling is done by growing the expression host in a growth media enriched with the desired isotopes. Since isotopically enriched compounds remain expensive, organisms are used which are capable of growing on a defined minimal medium, containing only one carbon-13 source, usually glucose, but occasionally glycerol or methanol, and one nitrogen-15 source such as ammonium chloride or ammonium sulfate . These organisms include organisms such as *Escherichia coli*, which is the most frequently used bacteria, and *Pichia pastoris*, which is the most frequently used yeast.The purified protein is usually dissolved in a buffer solution and adjusted to the desired solvent conditions.

Data collection

Protein NMR utilizes multidimensional nuclear magnetic resonance experiments to obtain information about the protein. Ideally, each distinct nucleus in the molecule experiences a distinct chemical environment and thus has a distinct chemical shift by which it can be recognized. However, in large molecules such as proteins the number of resonances can typically be several thousand and a one-dimensional spectrum inevitably has incidental overlaps. Therefore multidimensional experiments are performed which correlate the frequencies of distinct nuclei. The additional dimensions decrease the chance of overlap and have a

larger information content since they correlate signals from nuclei within a specific part of the molecule. Magnetization is transferred between nuclei using radiofrequency pulses and delays via one of many elaborate pulse sequences which allow connections between different nuclei to be detected. The array of nuclear magnetic resonance experiments used on proteins fall in two main categories — one where magnetization is transferred through the chemical bonds, and one where the transfer is through space, irrespective of the bonding structure. The first category is used to assign the different chemical shifts to a specific nucleus, and the second is primarily used to generate the distance restraints used in the structure calculation, and in the assignment with unlabelled protein.

Depending on the concentration of the sample, on the magnetic field of the spectrometer, and on the type of experiment, a single multidimensional nuclear magnetic resonance experiment on a protein sample may take hours or even several days to obtain suitable signal-to-noise ratio through signal averaging, and to allow for sufficient evolution of magnetization transfer through the various dimensions of the experiment. Other things being equal, higher-dimensional experiments will take longer than lower-dimensional experiments.

Typically the first experiment to be measured with an isotope-labelled protein is a 2D heteronuclear single quantum correlation (HSQC) spectrum where "heteronuclear" refers to nuclei other than 1H. In theory the heteronuclear single quantum correlation has one peak for each H bound to a heteronucleus. Thus in the 15N-HSQC one signal is expected for each amino acid residue with the exception of proline which has no amide-hydrogen due to the cyclic nature of its backbone. Tryptophan and certain other residues with N-containing sidechains also give rise to additional signals. The 15N-HSQC is often referred to as the fingerprint of a protein because each protein has a unique pattern of signal positions. Analysis of the 15N-HSQC allows researchers to evaluate whether the expected number of peaks is present and thus to identify possible problems due to multiple conformations or sample heterogeneity. The relatively quick heteronuclear single quantum correlation experiment helps determine the feasibility of doing subsequent longer, more expensive, and more elaborate experiments. It is not possible to assign peaks to specific atoms from the heteronuclear single quantum correlation alone.

Resonance assignment

In order to analyze the nuclear magnetic resonance data, it is important to get a resonance assignment for the protein. That is to find out which chemical shift in each dimension corresponds to which atom. Several different types of experiments have been invented to achieve this. The procedure depends on whether the protein is isotopically labelled or not, since a lot of the assignment experiments depend on carbon-13 and nitrogen-15.

With unlabelled protein the usual procedure is to record a set of two dimensional homonuclear nuclear magnetic resonance experiments through correlation spectroscopy (COSY), of which several types include conventional correlation spectroscopy, *total correlation* spectroscopy (TOCSY) and nuclear Overhauser effect spectroscopy (NOESY). A two-dimensional nuclear magnetic resonance experiment produces a two-dimensional spectrum. The units of both axes are chemical shifts. The COSY and TOCSY transfer magnetization through the chemical bonds between adjacent protons. The conventional correlation spectroscopy experiment is only able to transfer magnetization between protons on adjacent atoms, whereas in the total correlation spectroscopy experiment the protons are able to relay the magnetization, so it is transferred among all the protons that are connected by adjacent protons. Thus in a conventional correlation spectroscopy, an alpha proton transfers magnetization to the beta protons, the beta protons transfers to the alpha and gamma protons, if any are present, then the gamma proton transfers to the beta and the delta protons, and the process continues. In total correlation spectroscopy, the alpha and all the other protons are able to transfer magnetization to the beta, gamma, delta, epsilon if they are connected by a continuous chain of protons. The continuous chain of protons are the sidechain of the individual amino acids. Thus these two

experiments are used to build so called spin systems, that is build a list of resonances of the chemical shift of the peptide proton, the alpha protons and all the protons from each residue's sidechain. Which chemical shifts corresponds to which nuclei in the spin system is determined by the conventional correlation spectroscopy connectivities and the fact that different types of protons have characteristic chemical shifts. To connect the different spinsystems in a sequential order, the nuclear Overhauser effect spectroscopy experiment have to be used. Because this experiment transfers magnetization through space, it will show crosspeaks to all protons that are close in space regardless of whether they are in the same spin system or not. The neighbouring residues are inherently close in space, so the assignments can be made by the peaks in the NOESY with other spin systems.

One important problem using homonuclear nuclear magnetic resonance is overlap between peaks. This occurs when different protons have the same or very similar chemical shifts. This problem becomes greater as the protein becomes larger, so homonuclear nuclear magnetic resonance is usually restricted to small proteins or peptides.

The process of resonance assignment for a nitrogen-15 labelled sample is similar to the homonuclear case. No experiment can be performed that transfers magnetisation between two spin systems through bonds either. The main difference is the ability to record nitrogen-15 edited three dimensional experiments: TOCSY-N HSQC and NOESY-N-HSQC. These experiments build onto the HSQC experiment, but have an additional proton dimension. It can be visualised as each peak in the HSQC having the TOCSY or NOESY peaks stacked onto it. Thus if the TOCSY peak from an amide proton, HN, has a cross peak to its alpha proton, Halpha, at the coordinates (HN, Halpha) in the TOCSY spectrum, the corresponding peak would be at (HN, Halpha,N) in the TOCSY-N-HSQC. Thus it is possible to resolve overlaps in the proton dimension, if the corresponding nitrogens have chemical shifts distinct from one another.

Restraint generation

In order to make structure calculations a number of experimentially determined restraints have to be generated. These fall into different categories, the most widely used is distance restraints and angle restraints.

Distance restraints

A crosspeak in a NOESY experiment signifies spatial proximity between the two nuclei in question. Thus each peak can be converted in to a maximum distance between the nuclei, usually between 1,8 and 6 angstroms. The 30 intensity of a noesy peak is proportional to the distance to the minus 6th power, so the distance is determined according to intensity of the peak. The intensity-distance relationship is not exact, so usually a distance range is used.

It is of great importance to assign the noesy peaks to the correct nuclei based on the chemical shifts. If this task is performed manually it is usually very labor intensive, since proteins usually have thousands of noesy peaks. Some computer programs such as CYANA and ARIA/CNS perform this task automatically, coupled to a structure calculation. To obtain as accurate assignments as possible it is a great advantage to have access to carbon-13 and nitrogen-15 noesy experiments, since they help to resolve overlap in the proton dimension. This leads to faster and more reliable assignments, and in turn to better structures.

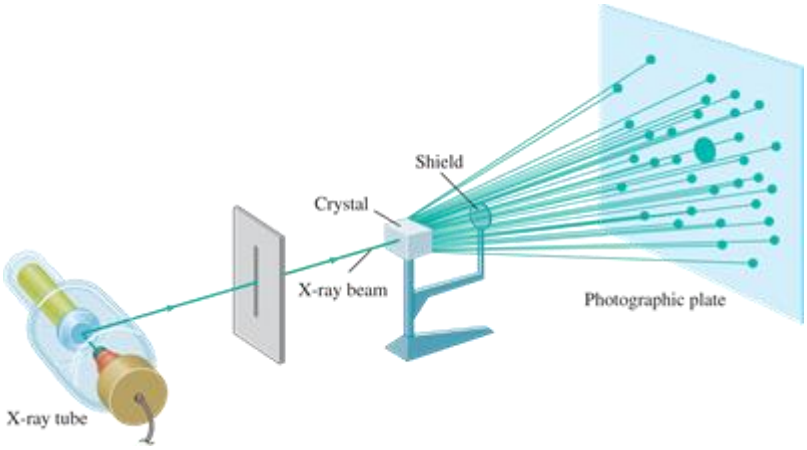
Angle restraints

In addition to distance restraints, restraints on the torsion angles of the chemical bonds, typically the psi and phi angles can be generated. One approach is to use the Karplus equation, to generate angle restraints from coupling constants. Another approach uses the chemical shifts to generate angle restraints. Both methods use the fact that the geometry around the alpha carbon affects the coupling constants and chemical shifts, so given the coupling constants or the chemical shifts, a qualified guess can be made about the torsion angles.

X-ray crystallography (X-ray diffraction)

X-ray crystallography is the science of determining the arrangement of atoms within a crystal using X-rays scattered from electrons. The oldest and most exact X-ray crystallographic method is *single-crystal X-ray diffraction*; for crystals of sufficient purity and regularity, this method gives the crystal structure, which defines the position of most atoms to within a few tenths of an Ångström. Since all atoms have electrons, X-ray crystallography may be used on any type of crystal, be they metals or minerals or engineered semiconductors or crystals of biomolecules such as aspirin or proteins. Such crystal structures have many scientific and medical applications; for example, they can account for unusual electronic or elastic properties of a material, shed light on the molecular events of enzyme catalysis or serve as the basis for structure-based drug design.

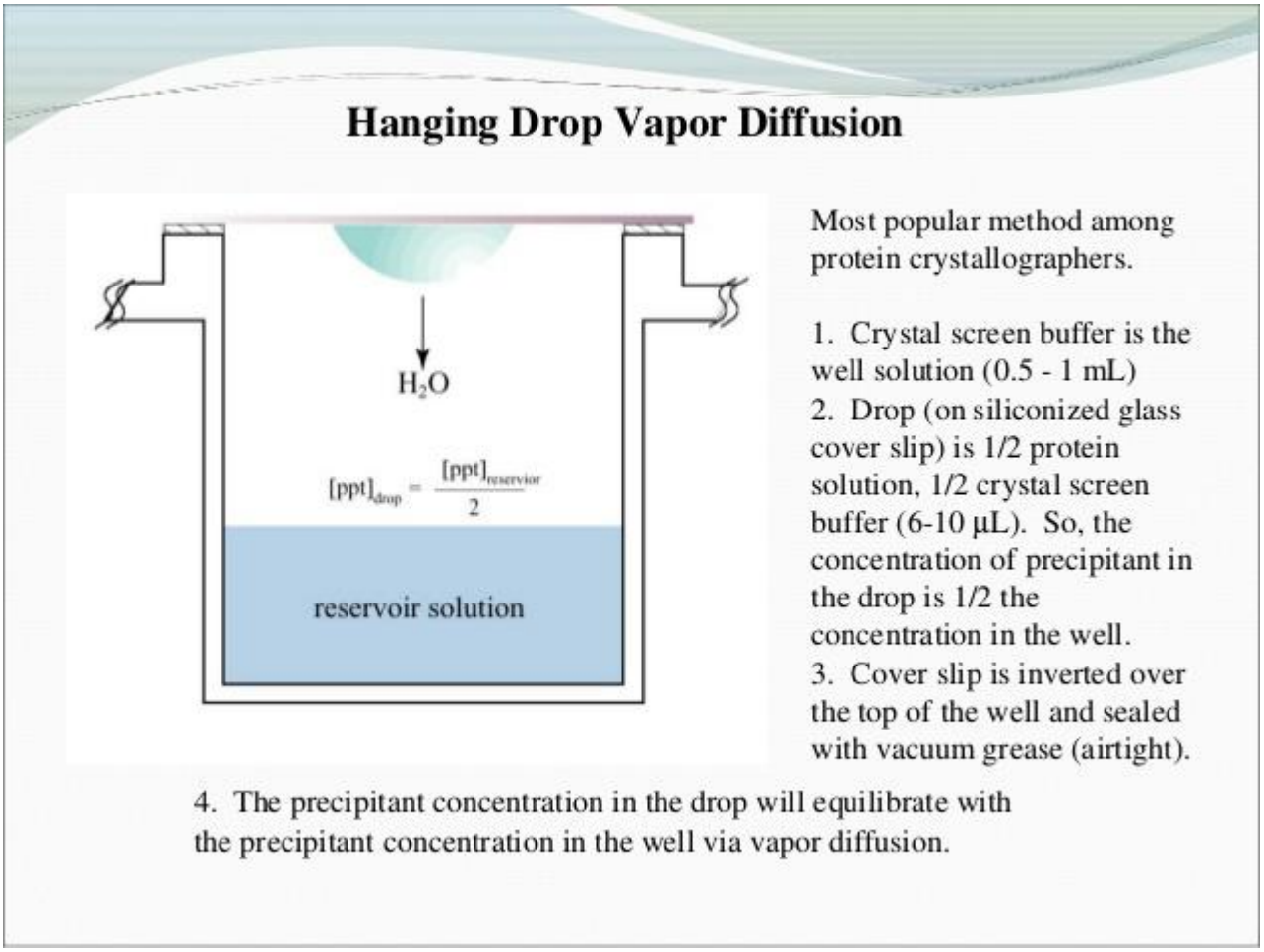
X-ray crystallography requires a crystal to amplify the signal; the periodicity of the electron density is used to diffract the X-rays with manageable measurement error. The technique involves the scattering of X-rays of a single wavelength ("monochromatic X-rays") from a single, pure crystal. This scattering produces a diffraction pattern, a set of intense spots (also called *reflections*) on a screen behind it. The spots can be related to the density of electrons in the crystal through a mathematical operation called a Fourier transform. This technique decomposes the electron density into its spatial frequency components, just as the human ear can discern different musical notes in a chord. Each spot observed on the screen corresponds to a spatial oscillation of the electron density along a particular direction within the crystal**.**



Protein crystallization

Most important part of any protein crystallography studies is protein crystallization. Without perfect crystals of protein (or any other biological samples) it is impossible to carry out any crystallographic structural studies. The aim of protein crystallization is to produce well-ordered protein mono-crystals without any inclusion and large enough to diffract X-Ray beam. Also you can find some techniques for low molecular weight peptide crystallization. Despite very wide knowledge about protein crystallization it is still impossible to predict any conditions for protein crystallization. Requirements for Protein crystallization

- Crystallization by Vapor Diffusion
- Crystallization by Dialysis
- Crystallization in space
- Automated Protein crystallization
- Crystallization conditions



Crystal mounting

Usually, protein crystals are grown in some solutions and before further analysis, protein crystals must be mounted and placed on the detector under the X-Ray beam. Protein Crystallography deals with crystals about 0.1 mm in size, that's why crystal mounting required a very high coordination of all motions and reasonable practice. Best advice is to spend 1-2 days to practice play with cheap protein crystals, for example Lysozyme. Currently a lot of efforts are targeted to automate crystal mounting process, but despite this the manual crystal mounting is a still very important part of protein crystallography. protein crystallography needs Crystal separating

- Crystal mounting in capillary
- Crystal mounting in the loop
- Crystals Freezing/refreezing
- Crystal storage in liquid nitrogen
- Automated crystal mounting
- Dry crystals

Data collection

X-ray Data collection is a central part of all protein crystallography. This chapter describes all modern equipment for X-ray data collection: detectors and X-ray radiation sources. Very important part of X-ray data collection is geometry of experiment. Correct alignment of the crystal sometimes allows improving completeness on the final data set. X-ray data collection is time consuming procedure, and this time is very expensive. It is possible to significantly minimize the time of X-ray data collection by correct prediction of

strategy of data collection. Multi-wavelength X-ray data collection, as a part of anomalous data collection will be discussed, as a most popular technique of crystal structure solution in protein crystallography.

Detector types

- Radiation damage
- Strategy prediction
- Anomalous diffraction
- Geometry of the data collection

Data processing

X-ray data processing is a key stage for protein crystallography. The final quality of X-Ray data is related with data processing procedure, which includes integration of the crystallographic data and scaling. In this chapter the basic programs for data processing will be discussed. You also can find manuals for most popular programs written by our specialists with set of samples of data processing for many different cases, from easy one to a very difficult. Also the role of some parameters widely used for data processing, such as integration spot radius or estimated error, will be discussed together with number of examples showing the impact of these parameters of data processing on final model and final statistic.

STRUCTURE-FUNCTION RELATIONSHIP

DNA-binding proteins

DNA-binding proteins are proteins that comprise any of many DNA-binding domains and thus have a specific or general affinity to DNA. DNA-binding proteins include transcription factors which modulate the process of transcription, nucleases which cleave DNA molecules, and histones which are involved in DNA packaging in the cell nucleus. Sequence-specific DNA-binding proteins generally interact with the major groove of B-DNA, because it exposes more functional groups that identify a base pair. The specificity of DNA-binding proteins can be identified using the DNase footprinting technique. DNA binding proteins consist of the zinc finger, the helix turn helix, and the leucine zipper (including other In addition to the core promoter region, genes transcribed by RNA polymerase II have regulatory sequences associated with them.

In multicellular eukaryotes these can be divided into promoter proximal elements, situated within 200 bp of the transcription start site, and enhancers, situated some distance upstream of the gene, or downstream, or even within the gene (in an intron).In yeast there is usually just one regulatory sequence situated upstream called the upstream activating sequence or UAS. DNA binding proteins bind to these regulatory sequences. Within the promoter proximal region there may be a number of binding sites and each enhancer may also have multiple binding sites. The yeast UAS usually has only one protein binding site.

Classes of DNA Binding Protein:
DNA-binding proteins can be divided into families according to their amino acid sequence:

- Helix-turn-helix

Homeodomain
POU domain

- Zinc finger

C2H2
C2C2

- Amphipathic helix
- Leucine zipper
- Helix-loop-helix

Helix-turn-helix proteins
These have a similar structure to bacterial regulatory proteins such as the repressor and cro proteins, the *lac* repressor and so on which bind as dimers and their binding sites are

palindromic. They contain 3 helical regions separated by short turns which are why they are called helix-turn-helix proteins. One protein helix (helix 3) in each subunit of the dimer occupies the major groove of two successive turns of the DNA helix.

Homeodomain proteins:

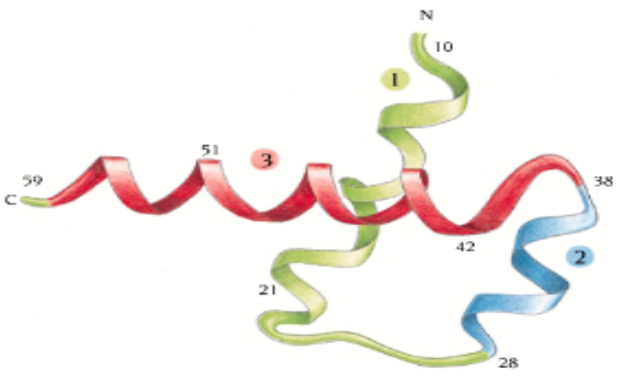
When the nucleotide sequences of a number of genes which control development of *Drosophila* were compared it was found that they share a common sequence of 180 base pairs with only a few differences between genes. Since the sequence was initially identified in a group of genes called homeotic genes the sequence was called the homeobox. The 180 bp corresponds to 60 amino acids in the corresponding protein. This protein domain is called the homeodomain. Homeodomain-containing proteins have since been identified in a wide range of organisms including vertebrates and plants. The homeodomain shows a high degree of sequence conservation. The homeodomain contains 4 helical regions. Helices II and III are connected by 3 amino acids comprising a turn. This region has a very similar structure to helices II and III of bacterial DNA binding proteins. There are some important differences between the homeodomain and prokaryotic helix-turn-helix proteins:

The homeodomain has an additional helix - helix number IV which may also play a part in binding to DNA. Homeodomain proteins bind as monomers rather than dimers and the binding sequences for homeodomain proteins are not palindromic.

Homeodomain Proteins

- involved in development
- DNA recognition motifs ~ 60 amino acids long

  e.g. Antennapedia in Drosophila, structure det'd by NMR
- 3 α-helices connected by short loops
- DNA-binding motif also HLH (HTH) as in Cro & repressor
- homeodomain recogⁿ α-helices longer, one end disordered unless bound to DNA



- homeodomain binds as monomer to 5' ATTA 3' with high affinity (~ 1 nM)
- again, recognition α_3 samples the major groove

(a) Has conserved residues

Asn51, Arg 53 } polar, interact with DNA

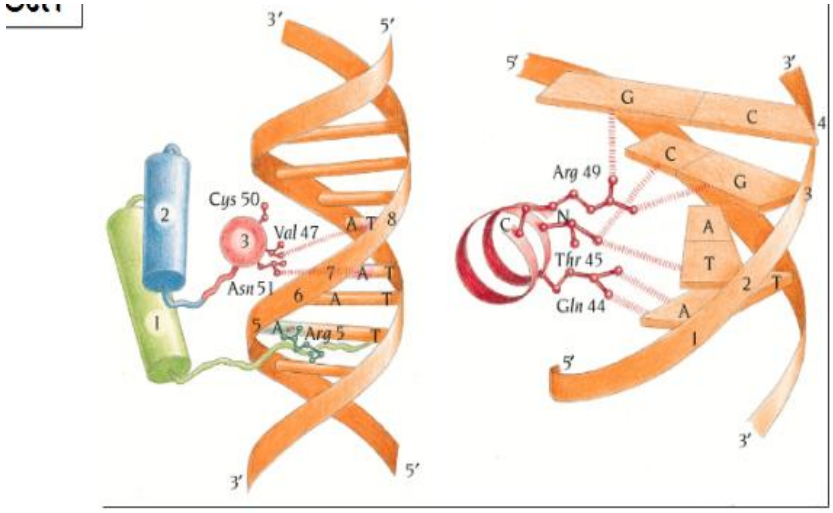
Trp 48, Phe49 } participate in hydrophobic core

(b) again, nonspecific H-bonds between DNA phosphates and Lys/Arg residues

Other quirk: lots of H-bonds in major grooves mediated by water molecules, B-DNA structure remains intact

POU proteins

These are helix-turn-helix DNA binding proteins, related to the homeodomain proteins. These proteins contain a homeodomain and next to the homeodomain there is a region of about 75 amino acids called the POU-specific domain or POU region. About 15-20 amino acids separate the 2 domains. The POU domain was first recognized when the sequence of a number of transcription factors was compared and a 75 amino acid region showed strong similarity between the different proteins. It was called the POU domain after the first letters of Pit-1, Oct-1 and Oct-2 and Unc-86. The POU structure has since been found in a number of other proteins. The homeodomain of POU proteins can bind to DNA on its own but only weakly. The presence of the POU domain is required for high affinity, sequence specific binding. POU proteins bind as monomers or dimers even though their binding site is not palindromic.



- both homeodomain & POU-specific domain have HLH motifs & recogⁿ helix inserted into major groove
- ↑ specificity, bind in tandem

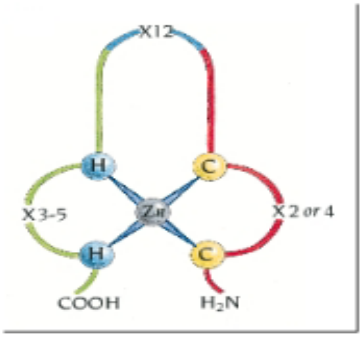
Zinc finger proteins

The C2H2 finger:

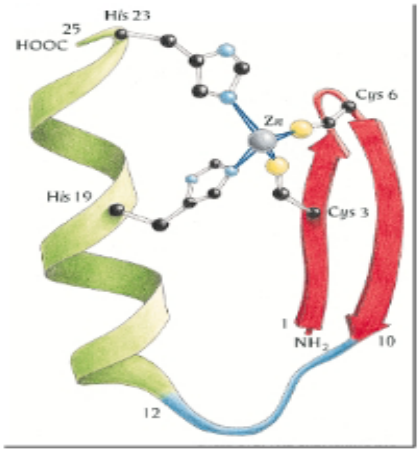
The first positively-acting eukaryotic regulatory protein to be well characterized was TFIIIA which is required for transcription of the 5S rRNA genes. When the amino acid sequence of this protein was determined it was found to have a 30 amino acid domain repeated 9 times. The general structure of these domains is: Phe (sometimes Tyr) - Cys - 2 to 4 amino acids - Cys - 3 amino acids - Phe (sometimes Tyr) - 5 amino acids - Leu - 2 amino acids - His - 3 amino acids - His. When the protein was purified it was found to have zinc associated with it and this zinc was shown to be required for the activity of the protein. Since cysteine and histidine were known to be the most important amino acids for binding zinc in proteins - a structure was proposed for the repeating units whereby repeated domains form loops so that the zinc is bound between pairs of cysteines and histidines - this is the zinc finger. The phenylalanine or tyrosine residues which occur at invariant positions are required for DNA binding. Similar sequences have been found in a range of other DNA binding proteins though the number of fingers varies. For example, the SP1 transcription factor which binds to the GC box found in the promoter proximal region of a number of genes has 3 fingers. This type of zinc finger which has 2 cysteines and 2 histidines is called a C2H2 zinc finger.

Zinc Fingers

- first observed in TFIIIA from Xenopus
- RNA pol III TF
- has ~9 repeats of ~30 aas each with 2 Cys & 2 His residues at approximately the same locations



- needs to coordinate Zn ion for stability
- miniglobular domain



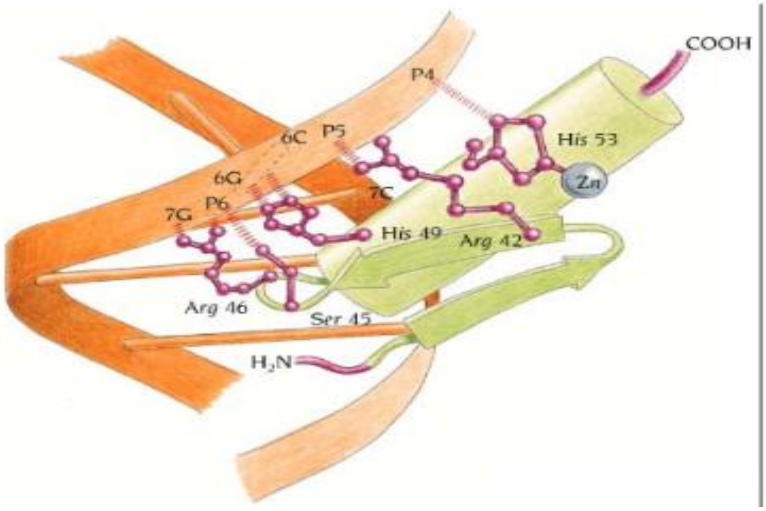
- tight binding but DNA remains undistorted
- each Zn-finger directly interacts with major groove

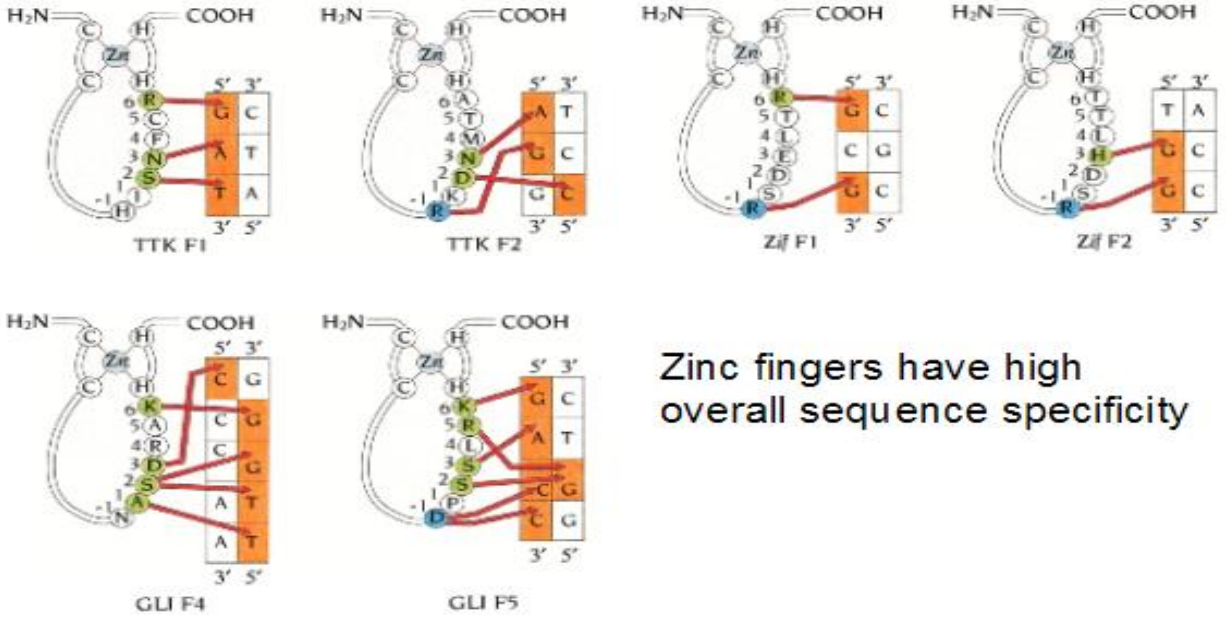
- N-terminus of helix interacts with bases

  Arg46 w/ G7

  His49 w/G6

- one strand of β-sheet interacts non-specifically with phosphate backbone





Zinc fingers have high overall sequence specificity

The C2C2 finger

Another type of zinc finger which binds zinc between 2 pairs of cysteines has been found in a range of DNA binding proteins. The general structure of this type of zinc finger is: Cys - 2 amino acids - Cys - 13 amino acids - Cys - 2 amino acids - Cys. This is called a C2C2 zinc finger.It is found in a group of proteins known as the steroid receptor super family, each of which has 2 C2C2 zinc fingers. Steroid hormone receptors all have the same mode of action: Firstly, the hormone enters the cell by diffusion. It then binds to the receptor protein, either in the cytoplasm or the nucleus. The receptor then enters the nucleus (if not already in it) and binds to a specific

DNA sequence close to a gene, activating transcription of that gene. DNA binding and transcriptional activation can only occur when the hormone is bound to the receptor. Steroid receptors bind to DNA as dimers and bind to palindromic DNA sequences called hormone response elements (HREs): Oestrogen receptor (ERE) GGTCAnnnTGACC Glucocorticoid receptor (GRE) GAACAnnnTGTTC Thyroid response element (TRE) GGTCATGACC The genes coding for a number of steroid hormone receptors have been cloned, sequenced and subjected to mutational analysis to find out which parts of the protein are important. The central region is required for DNA binding. On the C terminal side of this there is a region required for hormone binding, dimerisation and transcriptional activation. There is also an N terminal region required for transcriptional activation. The central DNA binding/dimerisation region contains two zinc fingers.If the three amino acids shown in red in the oestrogen receptor (EG--A) are changed to those found at the same positions in the glucocorticoid receptor (GS--V) then the receptor will bind to the GRE in response to oestrogen instead of to the ERE. The structure of the DNA-binding domain of the oestrogen and glucocorticoid receptors has been determined by 2D NMR. There are 2 helices which cross at right angles, one for each finger. There are 2 metal binding cysteines at the base of each helix. The other two lie in a curve of protein. In between there is a loop. The two recognition helices of a receptor dimer fit into the major groove of the DNA, pointing in The C2H2 and C2C2 zinc fingers have completely different structures: The helices are in different places - in the C2H2 finger the Zn binding histidines are at the C terminus end of the helix. In the C2C2 finger the Zn binding cysteines are at the N-terminal end of the helix. The C2C2 has no -sheet whereas the C2H2 does.

Leucine zipper proteins:

The first DNA binding protein of this type to be identified was purified from extracts of liver cells. It was called C/EBP because it is an enhancer binding protein and it was originally thought to bind to the CAAT promoter proximal sequence - though this is now known not to be the case. C/EBP will only bind to DNA as a dimer. The region of the protein where the two monomers join to make the dimer is called the dimerisation domain. This lies towards the C-terminal end of the protein. When the amino acid sequence was examined it was found that a leucine residue occur every seventh amino acid over a stretch of 35 amino acids. If this region were to form an helix then all of these Leucine would align on one face of the helix. One face of the helix is therefore very hydrophobic - since leucine has a hydrophobic side chain, whereas the opposite face has amino acids with charged side chains which are hydrophilic (amphipathic helix). If one of these leucines is changed to another amino acid dimerisation is prevented. It is proposed therefore that the two monomers bind together in the amphipathic helix region - hydrophobic interactions holding the monomers together. Adjacent to the leucine zipper region is a region of 20-30 amino acids which is rich in the basic (positively charged) amino acids lysine and arginine. This is the DNA binding domain - often referred to as the bZIP domain - the basic region of the leucine zipper.

Leucine zippers provide dimerization interactions for some eukaryotic transcription factors

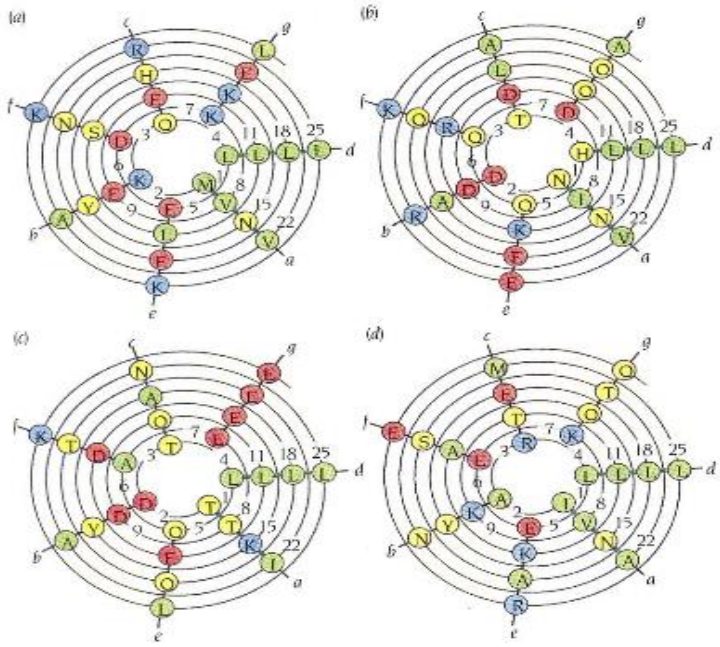


Figure 10.17 Amino acid sequences, represented as α helical wheels with 3.5 residues per turn, of a region of 28 residues from the DNA-binding domains of the transcription factors (a) GCN4, (b) Max, (c) Fos and (d) Jun. Positively charged residues are blue, negatively charged red; polar residues are yellow and hydrophobic, green. Residues with a spacing of seven residues are connected into spikes of the wheel which are labeled *a* to *g*. Side chains from each residue in a spike form a ridge on the outside of the α helix which is parallel to the helical axis. Spike *d* in all four sequences forms a heptad pattern of leucine residues that is characteristic of a leucine zipper coiled-coil α-helical structure.

Leucine zippers provide dimerization interactions for some eukaryotic transcription factors

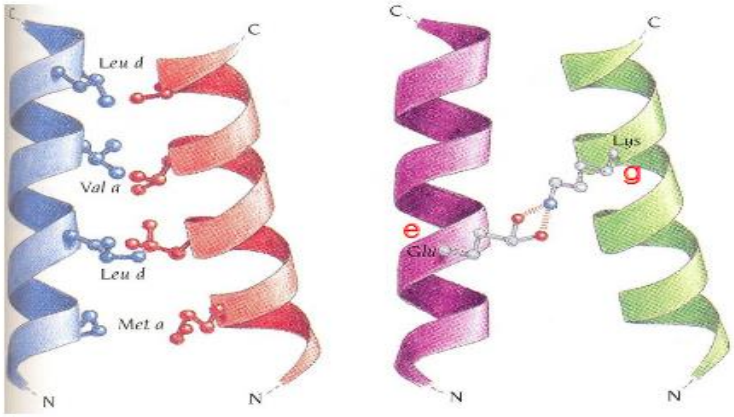


Figure 10.18 Side-chain interactions in the leucine zipper structure. (a) The hydrophobic side chains in spikes *a* and *d* (see Figure 10.17) form a hydrophobic core between the two coiled α helices. (b) Charged side chains in spikes *e* and *g* can promote dimer formation by forming complementary charge interactions between the two α helices.

Helix-loop-helix proteins

These are similar to leucine zippers in that they form dimers via amphipathic helices. They were first discovered as a class of proteins when a region of similarity was noticed between two enhancer binding proteins called E47 and E12. This conserved region has the potential to form two amphipathic helices (though without the repeated leucines) separated by a loop hence helix-loop-helix. Next to the dimerisation domain is a DNA binding domain, again rich in basic amino acids and referred to as the bHLH domain. These structures are also found in a number of genes required for development of the *Drosophila* nervous system - the *Achaete-scute* complex, and in a protein called MyoD required for mammalian muscle differentiation. All of these proteins can form homodimers or heterodimers with other members of the helix-loop-helix family. In some cases heterodimerisation has been shown to be a way of regulating the DNA binding activity.

Transcription factors

Nucleic acids are the storage of genetic information and this information has to be accessible and inherited. The regulation of gene expression and the replication of the genome are central mechanisms to provide a 'translation' of the genetic blueprint into the machinery of life - the proteins. Nucleic acids need proteins for storage, replication, and transcription purposes. The specificity of the transcription and replication requires recognition on the molecular level between protein structures and nucleic acid structures. The genetic code has to be readable. The reading of the code is a conformationally specific interaction between amino acids and nucleic acids. Surface properties of the macromolecules involved are the essential key in the recognition process. Electrostatic interaction, hydrogen bonding capability and hydrophobic effects are of importance. Complementarity in surface profiles is the essential mechanism that provides the specificity. Here we discuss a few selected protein systems that recognize DNA double helical structures. The sequence of the base pairs in the double helix specifies the local conformation of the double helix - its ribose-phosphate backbone and the dimensions of the minor and major grooves of the helix. Proteins that bind DNA and are involved in replication or transcription do so in a sequence specific way. Transcription factors are dimers when active, i.e., they bind to DNA upon dimerisation and are inactive in the monomeric form. Dimerisation is a regulatory mechanism of controlling transcription factor activity. There are 3 common features most DNA binding proteins have in common: - *Major groove* is the binding site of proteins through -helices; the dimensions of the major groove is 12Å wide and 8Å deep - *Minor groove* of B-DNA is 5Å wide, 8Å deep, and is generally too narrow to fit entire - -helices, but is recognized by -sheet structures of TATA box binding proteins - Sequence specific DNA binding proteins generally *do not disrupt the base pairs* of the DNA, but do distort backbone conformation by bending the double helix.

The HLH motif is involved in homodimer and heterodimer associations

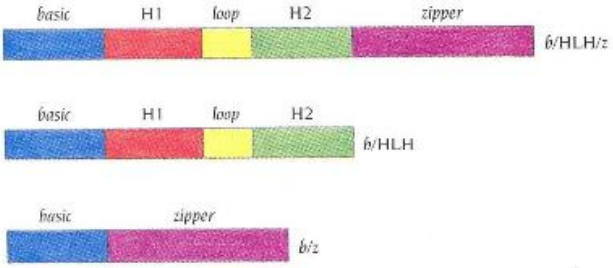


Figure 10.23 Domain arrangement along the polypeptide chains of three families of transcription factors; b/z , b/HLH and b/HLH/z. All three have a basic region (blue) that binds DNA. Dimerization is achieved by the zipper region (purple) in the b/z family, by the H1-loop-H2 region (red-yellow-green) in the b/HLH family and by a combination of both the zipper and the HLH regions in the b/HLH/z family.

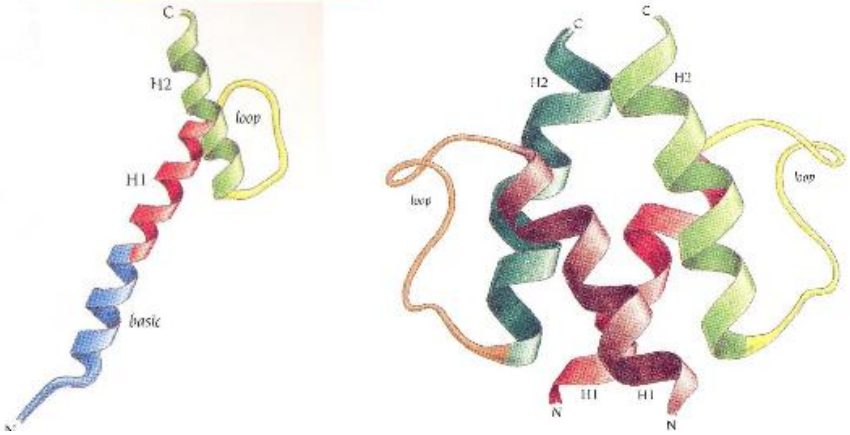


Figure 10.28 Structure of the dimerization region of MyoD. The α helices H1 (red and brown) and H2 (light and dark green) of the two monomers form a four-helix bundle that keeps the dimer together. The loops (yellow and orange) are on the outside of the four-helix bundle. (Adapted from P.C.M. Ma et al., *Cell* 77: 451–459, 1994.)

The α-helical basic region of the b/HLH motif binds in the major groove of DNA

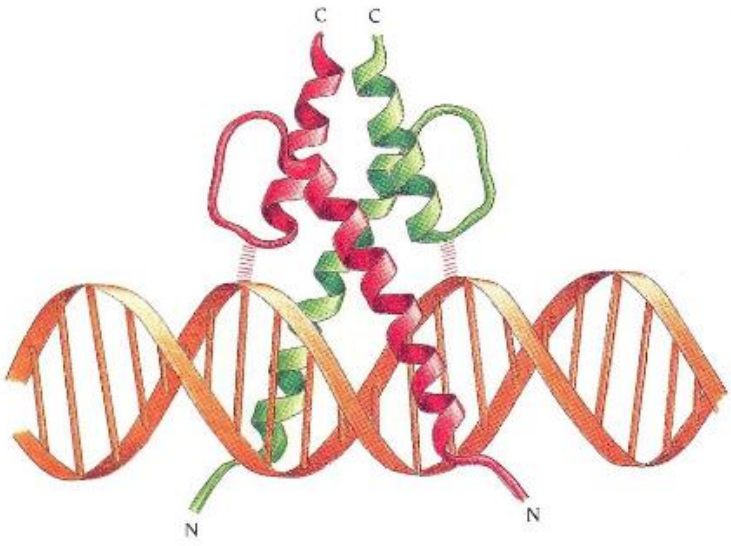


Figure 10.26 Schematic diagram of the binding of MyoD to DNA. MyoD binds as a dimer (red and green) with the N-terminal basic regions interacting with the major groove of DNA (orange). The monomers diverge from each other after the H1 region and bind on opposite sides of DNA, as for GCN4 (see Figure 10.21). The four-helix bundle, together with contacts with phosphates of the DNA backbone, rigidify the fork and allow MyoD only to bind to enhancer elements with a specific spacing between the half-sites. The DNA structure is essentially B-DNA. (Adapted from P.C.M. Ma et al., *Cell* 77: 451–459, 1994.)

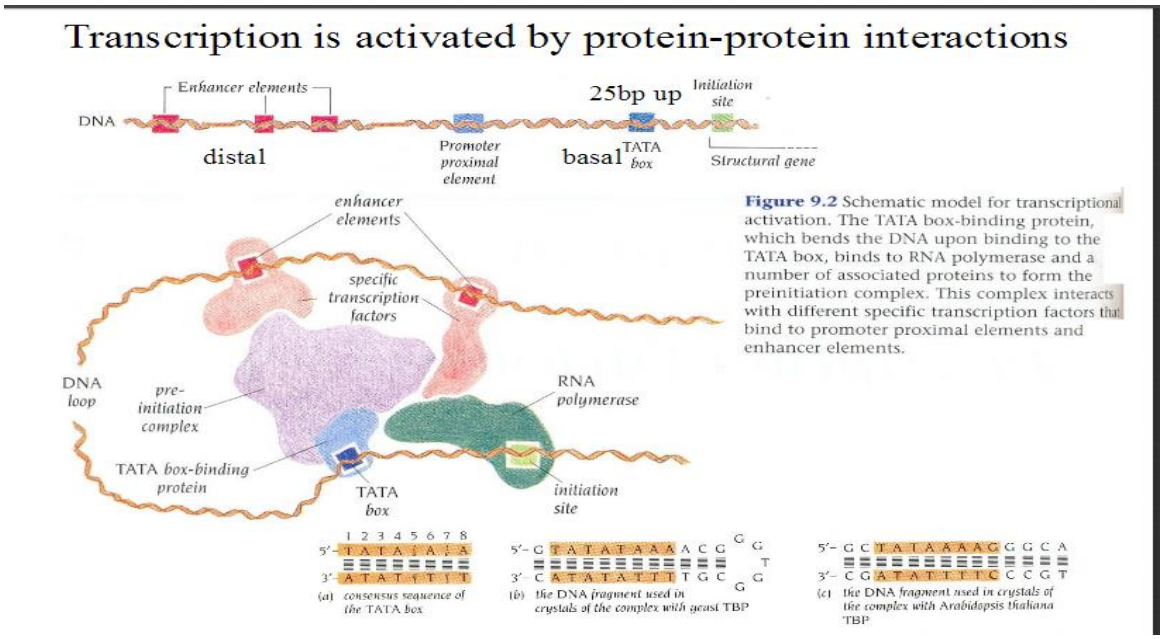


Figure 9.2 Schematic model for transcriptional activation. The TATA box-binding protein, which bends the DNA upon binding to the TATA box, binds to RNA polymerase and a number of associated proteins to form the preinitiation complex. This complex interacts with different specific transcription factors that bind to promoter proximal elements and enhancer elements.

Helix-turn-helix motif (prokaryotic systems)

The helix-turn-helix motif is the common DNA recognition motif in prokaryotes. The motif resembles that of an EF hand described in calmodulin. The F-helix is the recognition helix and the side chains give the specificity of binding. Sometimes more than one protein competes for the same sequence. The examples are bacteriophages P22, and 434, where repressors and activators affect transcription. They can recognize the same DNA fragment. The phage 434 *cI repressor* and *cro* have been shown *in vitro* to bind to the same 20bp DNA fragment. They have different binding interaction visualized by the two close, but not identical structures as determined by X-ray crystallography. They differ in their affinity for the same sequence, or DNA conformation, respectively through H-bonds, salt bridges and Van der Waals interactions. The *relative concentrations* of all proteins, therefore, determine which one is bound to the binding element most of the time, which in turn affects polymerase binding to the DNA determining if and in which direction transcription will occur. The ratio of all DNA binding proteins therefore determines the rate of transcription controlled by the DNA sequences in question. TRP Repressor: The trp repressor is a biologically interesting molecule. It is involved in the regulation of transcription of the genes for the enzymes that *E. coli* need to make the amino acid tryptophan. The trp repressor regulates the level of tryptophan through negative control. When the concentration of tryptophan is high, the products of the genes are not needed and the repressor binds to the operon region of the genes, shutting off transcription. The repressor is made as an aporepressor by a gene upstream of the trp operon. The aporepressor alone cannot bind to the operon of the trp genes. Tryptophan must be bound to the aporepressor. Tryptophan is a hydrophobic amino acid that occupies a hydrophobic pocket in the aporepressor. When the

tryptophan occupies the pocket, the shape of the protein is changed. This changes the conformation of the alpha helix that recognizes and binds to the DNA, allowing it to hydrogen bond with the operator. The binding between the repressor and the ligand is not direct. The binding is facilitated by water through hydrogen bonds. This changes the conformation, allowing the repressor to bind to the DNA and block transcription of the trp genes. By rotating the molecule it will be evident that there are two protein molecules bound to the DNA. This is due to the fact that the trp repressor must bind to the DNA as a dimer in order to be effective. Each monomer of the repressor must have a tryptophan molecule bound to it.

Tryptophan repressor is a dimeric protein (2 x 12,500) which regulates transcription of at least 5 genes; When the amino acid tryptophan is plentiful in the cell, it binds to the protein, which then binds in turn to these 5 DNA sequences, shutting off the genes. One of these genes, trpR, codes for the production of tryptophan itself and the other code for the tryptophan repressor protein. These are forms of feedback regulation. The trp (tryptophan) repressor is a 25 KD protein which regulates transcription of the tryptophan biosynthetic pathway in bacteria. There are 5 operons which are regulated by trpR: the *trpEDCBA*, *trpR*, *AroH*, *AroL* and *mtr* operons.

Mechanism:

When the amino acid tryptophan is in plentiful supply in the cell, trpR binds 2 molecules of tryptophan, which alters its structure and dynamics so that it becomes able to bind to operator DNA. When this occurs, transcription of the DNA is prevented, suppressing the products of the gene - proteins which make more tryptophan. When the cellular levels of tryptophan decline, the tryptophan molecules on the repressor fall off, allowing the repressor to return to its inactive form. TrpR also controls the regulation of its own production, through regulation of the *trpR* gene (Kelley & Yanovsky, 1982 PNAS USA 79 3120-3124).

The structure of the ligand bound holorepressor, and the ligand-free forms have been determined by both X-ray crystallography and NMR. The trp operon consists of a regulatory gene, a promoter, an operator, and a terminator. The trp operon only works in the presence of tryptophan. If there isn't enough tryptophan, the repressor protein breaks off from the operator (where the repressor is normally bound) and RNA polymerase can complete its reading of the strand of DNA. If the RNA polymerase reaches the terminator (at the end of the DNA strand), the enzyme for tryptophan is made.

Trp Repressor and DNA: Structurally, trp-repressor is a symmetric dimer, a molecule formed of two identical subunits (light blue and pink) intertwined with each other. The red atoms represent the two tryptophan ligands, molecules that bind with the represssor before it binds with DNA (light green). Calculations at the Pittsburgh Supercomputing Center (conducted by Daqing Zhao) showed that the tryptophan ligands stabilize the otherwise more flexible DNA-binding region of the protein and hold it at an optimal configuration for binding with DNA. The repressor regions directly above and below the tryptophan are the protein's "reading heads." They make direct contact with the DNA, wedging into DNA's "major groove" -- the opening that forms between DNA's two helical backbones. The DNA, in turn, bends at each end to wrap around the protein.

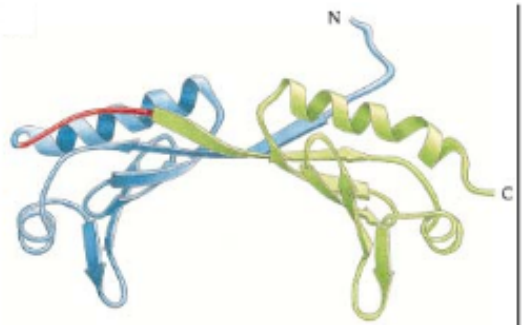
EUKARYOTIC TRANSCRIPTION: The central dogma of molecular biology stipulates that DNA gives rise to RNA which in turn gives rise to protein. It follows that a primary process in the molecular biology of the cell is production of RNA from DNA, i.e. the process of transcription. The failure of this process to occur, or to decode faithfully the information encoded in the DNA, would render the cell non-viable. In most eukaryotic cells, most genes are regulated (during development and differentiation, in response to specific cellular signals), at least in part, at the level of transcription initiation. Transcription and its regulation are fundamentally essential processes in the cell. RNA polymerases: The process of transcription involves polymerisation of nucleoside triphosphates into RNA, in a DNA template-dependent manner, and in eukaryotes it is catalysed by three RNA polymerase enzymes, RNA polymerase I, RNA polymerase II and RNA polymerase III. RNA polymerase I is the most prominent polymerase activity in the cell, followed by polymerase II while polymerase III is the smallest polymerase activity. RNA polymerases are huge multi-component complexes of over 10 protein subunits and are around 500 KDa in size. There is considerable relatedness between the three eukaryotic RNA polymerases and to the prokaryotic *E.Coli* RNA polymerase, especially between the largest and second largest subunits. RNA polymerase II differs from the others in that the largest subunit has a carboxy terminal extension called the carboxy terminal domain (CTD). The CTD contains a highly repeated heptapeptide (Tyr-Ser-Pro-Thr-Ser-Pro-Ser) which can be heavily phosphorylated. This phosphorylated domain is essential for transcription by RNA polymerase II in most eukaryotes, and it also links the processes of transcription and RNA processing. Initiation, elongation and termination:

RNA polymerases alone cannot recognize an initiation site in the genome. This requires the ordered formation of a stable transcription initiation complex composed of the RNA polymerase complex and another protein complex of a similar size, the pre-initiation complex, consisting of a number of general transcription factors (see below). In addition, experiments in yeast have shown that further “Mediator complex” associates with the CTD of RNA polymerase II and this 60-70 protein complex, termed the holoenzyme complex binds to the promoter and locates the polymerase to the gene's initiation site. At this stage the CTD is unphosphorylated, however upon transcription initiation the holoenzyme complex dissociates in part, the CTD of the RNA polymerase becomes highly phosphorylated and some new transcription factors associate with it. This new elongation complex transcribes RNA processively along the gene at a rate of up to 2000 nucleotides per minute. At the 3' end of the gene the elongation complex transcribes through the polyadenylation site and terminates at some point downstream where it disassociates from the DNA. There are defined termination signals for RNA polymerases I and III but the site(s) and mechanism of termination for RNA polymerase II has not yet been worked out. However, it is now clear that termination is linked with formation of the 3' end of the transcript via the CTD. The process of termination is important because it avoids the situation of the elongation complex travelling into the promoter of the next gene downstream, which would probably result in decreased transcription initiation for that gene. The basic transcription machinery TBP or TATA binding protein: One key general transcription factor common to all 3

RNA polymerase complexes, is TATA binding protein or TBP. TBP is a 38 KDa saddle-shaped monomer which can contact and bend severely TATA-containing DNA in the minor groove. TBP therefore changes the conformation of the DNA and this is thought to facilitate transcription factor binding. It is highly conserved from yeast to mammals and is evolutionarily ancient, a related protein being found in archaebacteria. TBP presents a wide outer surface for simultaneous binding of a number of TBP-associated factors (TAFs) and the complexes these form are the positioning factors for RNA polymerase. TAFs appear to regulate the activity of TBP and together they determine the specificity of polymerase binding to promoters.

- TATA-Box binding prns (TBPs) from yeast and Arabidopsis were cloned, overexpressed in *E. coli* & purified
- cocrystallized with a small segment of DNA representing TATA consensus
- X-ray structures of TBP from both species very similar, expect all TBPs to be closely related
- structure exhibits quasi-2-fold symmetry, 2 very similar domains connected by a loop

Saddle shape



TRANSMEMBRANE PROTEINS

The plasma membrane serves as the interface between the machinery in the interior of the cell and the extracellular fluid (ECF) that bathes all cells. The lipids in the plasma membrane are chiefly phospholipids like phosphatidyl ethanolamine and cholesterol. Phospholipids are amphiphilic with the hydrocarbon tail of the molecule being hydrophobic; its polar head hydrophilic. As the plasma membrane faces watery solutions on both sides, its phospholipids accommodate this by forming a **phospholipid bilayer** with the hydrophobic tails facing each other.

Transmembrane proteins are amphipathic, in that they have hydrophobic and hydrophilic regions that are oriented in the same regions in the lipid bilayer. Another name for them is **"integral proteins"**. Other types of proteins may be linked only at the cytoplasmic surface (by

attachment to a fatty acid chain), or at the external cell surface, attached by a oligosaccharide. Or, these non-transmembrane proteins may be bound to other membrane proteins. Collectively these are called **"peripheral membrane proteins".** Proteins inserted once through the membrane are called "single-pass transmembrane proteins." Those that pass through several times are called "multipass transmembrane proteins". They form loops outside the membrane.

Peripheral Membrane Proteins:

These are more loosely associated with the membrane. They are usually attached noncovalently to the protruding portions of integral membrane proteins. **Membrane proteins are often restricted in their movements:** A lipid bilayer is really a film of oil. Thus we might expect that structures immersed in it would be relatively free to float about. For some membrane proteins, this is the case. For others, however, their mobility is limited:

Some of the proteins exposed at the interior face of the plasma membrane are tethered to cytoskeletal elements like actin microfilaments.

Some proteins are the exterior face of the plasma membrane is anchored to components of the extracellular matrix like collagen.

Integral membrane proteins cannot pass through the tight junctions found between some kinds of cells (e.g., epithelial cells).

Prediction of Transmembrane regions in proteins:

PRED-TMR is a method that predicts transmembrane domains in proteins using solely information contained in the sequence itself. The algorithm refines a standard hydrophobicity analysis with a detection of potential termini ("edges", starts and ends) of transmembrane regions. This allows both to discard highly hydrophobic regions not delimited by clear start and end configurations and to confirm putative transmembrane segments not distinguishable by their hydrophobic composition.

Applied on several test sets of transmembrane proteins, the system gives a perfect prediction rating of 100% by classifying all the sequences in the transmembrane class. Applied on 995 non-transmembrane protein extracted from the PDBSELECT database, the neural network predicts falsely 23 of them to be transmembrane (97.7% of correct assignment).

BACTERIORHODOPSIN:

Bacteriorhodopsin is a photosynthetic pigment used by archaea, most notably halobacteria. It acts as a proton pump, i.e. it captures light energy and uses it to move protons across the membrane out of the cell. The resulting proton gradient is subsequently converted into chemical energy. Bacteriorhodopsin is an integral membrane protein usually found in two-dimensional crystalline patches known as "purple membrane", which can occupy up to nearly 50% of the surface area of the archaeal cell. The repeating element of the hexagonal lattice is composed of three identical protein chains, each rotated by 120 degrees relative to the others. Each chain has

seven transmembrane alpha helices and contains one molecule of retinal buried deep within, the typical structure for retinylidene proteins.

Conformational change:

It is the retinal molecule that changes its conformation when absorbing a photon, resulting in a conformational change of the surrounding protein and the proton pumping action. The bacteriorhodopsin molecule is purple and is most efficient at absorbing green light (wavelength 500-650 nm, with the absorption maximum at 568 nm).The three-dimensional tertiary structure of bacteriorhodopsin resembles that of vertebrate rhodopsins, the pigments that sense light in the retina. Rhodopsins also contain retinal, however the functions of rhodopsin and bacteriorhodopsin are different and there is no homology of their amino acid sequences. Both rhodopsin and bacteriorhodopsin belong to the 7TM receptor family of proteins, but rhodopsin is a G protein coupled receptor and bacteriorhodopsin is not. In the first use of electron crystallography to obtain an atomic-level protein structure, the structure of bacteriorhodopsin was resolved in 1990. It was then used as a template to build models of other G protein-coupled receptors before crystallographic structures were also available for these proteins.

Many molecules have homology to bacteriorhodopsin, including the light-driven chloride pump halorhodopsin (for whom the crystal structure is also known), and some directly light-activated channels like channelrhodopsin. All other photosynthetic systems in bacteria, algae and plants use chlorophylls or bacteriochlorophylls rather than bacteriorhodopsin. These also produce a proton gradient, but in a quite different and more indirect way involving an electron transfer chain consisting of several other proteins. Furthermore, chlorophylls are aided in capturing light energy by other pigments known as "antennas"; these are not present in bacteriorhodopsin based systems.

Photosynthetic reaction centre:

A photosynthetic reaction centre is a protein that is the site of the light reactions of photosynthesis. The reaction centre contains pigments such as chlorophyll and phaeophytin. These absorb light, promoting an electron to a higher energy level within the pigment. The free energy created is used to reduce an electron acceptor, and is critical for the production of chemical energy during photosynthesis.

Reaction centres are present in all green plants and in many bacteria and algae. Green plants have two reaction centres known as photosystem I and photosystem II and the structures of these centres are complex, involving a multisubunit protein. The reaction centre found in *Rhodopseudomonas* bacteria is currently better understood since it has fewer proteins than the examples in green plants.

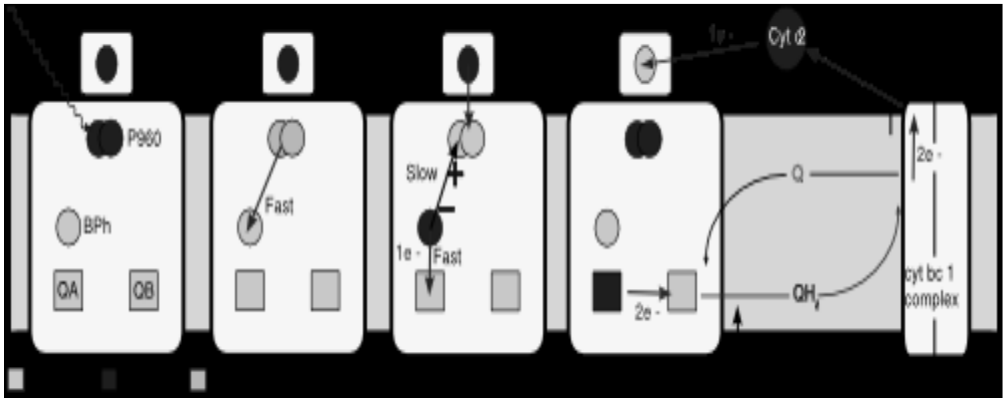
Capturing light energy:

A reaction centre is laid out in such a way that it captures the energy of a photon using pigment molecules and turns it into a usable form. Once the light energy has been absorbed directly by the pigment molecules, or passed to them by resonance transfer from antenna pigments, they release two electrons into an electron transport chain.

Light is made up of small bundles of energy called photons. If a photon with the right amount of energy hits an electron it will raise the electron to a higher energy level. Electrons are most stable at their lowest energy level or ground state, the orbit in which the electron has the least amount of energy. Electrons in higher energy levels can return to ground state in a manner analogous to a ball falling down a staircase. In doing so they release energy. This is the process which is exploited by a photosynthetic reaction centre. When an electron rises to a higher energy level it increases the reduction potential of the molecule. This means it has a greater tendency to donate electrons, the key to the conversion of light energy to chemical energy. In green plants, the electron transport chain that follows has many electron acceptors including phaeophytin, quinone, plastoquinone, cytochrome bf, and ferredoxin that ultimately result in the reduced molecule NADPH. The passage of the electron through the electron transport chain also results in the pumping of protons (hydrogen ions) from the chlorplast's stroma into the lumen resulting in a proton gradient across the thylakoid membrane that can be used to synthesis ATP using ATP synthase. Both the ATP and NADPH are used in the Calvin cycle to fix carbon dioxide into triose sugars.

Bacteria : The bacterial photosynthetic reaction centre has been an important model to understand the structure and chemistry of the biological process of capturing light energy. In the 1960s, Roderick Clayton was the first to purify the reaction centre complex from purple bacteria. However, the first crystal structure was determined by Hartmut Michel, Johann Deisenhofer and Robert Huber for which they shared the Nobel Prize in 1988. This was also significant since it was the first structure for any membrane protein complex.

fig-Bacterial photosynthetic reaction centre.



Four different subunits were found to be important for the function of the photosynthetic reaction centre. The L and M subunits, shown in blue and purple in the image of the structure both span the plasma membrane. They are structurally similar to one another, both having 5 transmembrane polypeptide helices. Four bacteriochlorophyll b (BChl-b) molecules, two bacteriophaeophytin b molecules (BPh) molecules, two quinones (QA and QB), and a ferrous ion

are associated with the L and M subunits. The H subunit, shown in gold, lies on the cytoplasmic side of the plasma membrane. A cytochrome subunit, shown in green, contains four c-type hemes and is located on the periplasmic surface (outer) of the membrane. The reaction centre contains two pigments that serve to collect and transfer the energy from photon absorption: BChb and Bph. BChb roughly resembles the chlorophyll molecule found in green plants, but due to minor structural differences, its peak absorption wavelength is shifted into the infrared, with wavelengths as long as 1000nm. Bph has the same structure as BChb, but the central magnesium ion is replaced by two protons.

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ENZYMES

SERINE PROTEASE:

Over one third of all known proteolytic enzymes are serine proteases. This large family of enzymes hydrolyzes the peptide bond of substrate via a nucleophilic serine residue in the active site, Ser-195 (chymotrypsin numbering), arranged with Asp-102 and His-57 into a catalytic triad (Hedstrom, 2002). Barrett and Rawlings have devised a classification scheme based on statistically significant similarities in sequence and structure of all known proteolytic enzymes and terms this database MEROPS (Rawlings et al., 2006). This classification system divides proteases into clans based on catalytic mechanism and families on the basis of common ancestry. At present, over 48,000 peptidase gene sequences have been classified into 50 clans and 184 families. Over 18,000 serine proteases are grouped into 12 clans and 40 families. X-ray crystal structures have been determined for a few representatives of each family. Serine proteases are widely distributed in nature and found in all kingdoms of cellular life including viral genomes. Clan PA peptidases bearing the chymotrypsin-fold are the largest and best studied family of serine proteases. They are abundantly represented in vertebrate organisms, but rare constituents of plant genomes. Prototypic members of this clan are trypsin and chymotrypsin, which cleave polypeptide chains on the C-terminal side of a positively charged side-chain (Arg or Lys) or large hydrophobic residue (Phe, Trp, Tyr), respectively. Initiation of protease activity is largely controlled by zymogen activation. Cessation of protease activity in vivo is achieved by one of three classes of protease inhibitors - canonical, non-canonical and serpin. In the human genome, protease genes exceeds >5-fold protease inhibitor genes, suggesting that each inhibitor likely regulates multiple proteases. Serine proteases play diverse roles in human health, from non-specific digestion to highly regulated functions like embryonic development, immune response and blood coagulation. Such processes require timely proteolytic activity, or highly regulated and specific limited proteolysis. In many members of the family, activity and specificity are allosterically regulated by macromolecular ligands, or small cations like Na^+ and Ca^{2+} (Page & Di Cera, 2006). Insufficient or excess protease activity can promote significant pathologies like cancer, inflammation, hemophilia, heart attack, stroke, and pancreatitis and parasite infection. Notably, genetic deficiencies of protease inhibitors are also associated with disease states. Industrial usage of proteolytic enzymes is widespread and commercially important. Production of subtilisin, a clan SB peptidase, for use in detergents is on the scale of tons/year and accounts for 40% of enzyme sales worldwide. Other industrial applications of proteases include the production of foodstuffs, leather, pharmaceuticals,

diagnostic reagents, waste management, and silver recovery.

In biochemistry, **serine proteases** or **serine endopeptidases** (newer name) are a class of peptidases (enzymes that cleave peptide bonds in proteins) that are characterised by the presence of a serine residue in the active site of the enzyme. Serine proteases are grouped into clans that share structural homology and then further subgrouped into families that share close sequence homology. The major clans found in humans include the chymotrypsin-like, the subtilisin-like, the alpha/beta hydrolase, and signal peptidase clans. Serine proteases participate in a wide range of functions in the body, including blood clotting, immunity, and inflammation, as well as contributing to digestive enzymes in both prokaryotes and eukaryotes.

Chymotrypsin-clan:

The three serine proteases of the chymotrypsin-like clan that have been studied in greatest detail are chymotrypsin, trypsin, and elastase. All three enzymes are synthesized by the pancreatic acinar cells, secreted in the small intestine and are responsible for catalyzing the hydrolysis of peptide bonds. All three of these enzymes are similar in structure, as shown through their X-ray structures. The differing aspect lies in the peptide bond which is being cleaved, which is called the scissile bond. The different enzymes, like most enzymes, are highly specific in the reactions they catalyze. Each of these digestive serine proteases targets different regions of a polypeptide chain, based upon the side chains of the amino acid residues surrounding the site of cleavage:

Chymotrypsin is responsible for cleaving peptide bonds following a *bulky hydrophobic* amino acid residue. Preferred residues include phenylalanine, tryptophan and tyrosine, which fit into a snug hydrophobic pocket.

Trypsin is responsible for cleaving peptide bonds following a *positively-charged* amino acid residue. Instead of having the hydrophobic pocket of the *chymotrypsin*, there exists an aspartic acid residue at the base of the pocket. This can then interact with positively-charged residues such as arginine and lysine on the substrate peptide to be cleaved.

Elastase is responsible for cleaving peptide bonds following a *small neutral* amino acid residue, such as Alanine, glycine and valine. (These amino acid residues form much of the connective tissues in meat). The pocket that is in "trypsin" and "chymotrypsin" is now partially filled with valine and threonine, rendering it a mere depression, which can accommodate these smaller amino acid residues. The combination of these three enzymes make an incredibly effective digestive team, and are primarily responsible for the digestion of proteins.

Subtilisin:

Subtilisin is a serine protease in prokaryotes. Subtilisin is evolutionary unrelated to the chymotrypsin-clan, but shares the same catalytic mechanism utilising a catalytic triad, to create a nucleophilic serine. This is the classic example used to illustrate convergent evolution, since the same mechanism evolved twice independently during evolution.

Catalytic mechanism:

The main player in the catalytic mechanism in the chymotrypsin and subtillisin clan enzymes mentioned above is the *catalytic triad*. The triad is located in the active site of the enzyme, where catalysis occurs, and is preserved in all serine protease enzymes. The triad is a coordinated structure

consisting of three essential amino acids: histidine (His 57), serine (Ser 195) (hence the name "serine protease") and aspartic acid (Asp 102). Located very near one another near the heart of the enzyme, these three key amino acids each play an essential role in the cleaving ability of the proteases.
In the event of catalysis, an ordered mechanism occurs in which several intermediates are generated. The catalysis of the peptide cleavage can be seen as a ping-pong catalysis, in which a substrate binds (in this case, the polypeptide being cleaved), a product is released (the N-terminus "half" of the peptide), another substrate binds (in this case, water), and another product is released (the C-terminus "half" of the peptide). Each amino acid in the triad performs a specific task in this process:
*The serine has an -OH group that is able to act as a nucleophile, attacking the carbonyl carbon of the scissile peptide bond of the substrate.
A pair of electrons on the histidine nitrogen has the ability to accept the hydrogen from the serine -OH group, thus coordinating the attack of the peptide bond.
The carboxyl group on the aspartic acid in turn hydrogen bonds with the histidine, making the pair of electrons mentioned above much more electronegative.

The whole reaction can be summarized as follows:
The polypeptide substrate binds to the surface of the serine protease enzyme such that scissile bond is inserted into the active site of the enzyme, with the carbonyl carbon of this bond positioned near the nucleophilic serine.
The serine -OH attacks the carbonyl carbon, and the nitrogen of the histidine accepts the hydrogen from the -OH of the [serine] and a pair of electrons from the double bond of the carbonyl oxygen moves to the oxygen. As a result, a tetrahedral intermediate is generated.

The bond joining the nitrogen and the carbon in the peptide bond is now broken. The covalent electrons creating this bond move to attack the hydrogen of the histidine, breaking the connection. The electrons that previously moved from the carbonyl oxygen double bond move back from the negative oxygen to recreate the bond, generating an acyl-enzyme intermediate.
Now, water comes in to the reaction. Water replaces the N-terminus of the cleaved peptide, and attacks the carbonyl carbon. Once again, the electrons from the double bond move to the oxygen making it negative, as the bond between the oxygen of the water and the carbon is formed. This is coordinated by the nitrogen of the histidine. which accepts a proton from the water. Overall, this generates another tetrahedral intermediate.
In a final reaction, the bond formed in the first step between the serine and the carbonyl carbon moves to attack the hydrogen that the histidine just acquired. The now electron-deficient carbonyl carbon re-forms the double bond with the oxygen. As a result, the C-terminus of the peptide is now ejected.

Additional stabilizing effects:
It was discovered that additional amino acids of the protease, *Gly 193* and *Ser 195*, are involved in creating what is called an *oxyanion hole*. Both *Gly 193* and *Ser 195* can donate backbone hydrogens

for hydrogen bonding. When the tetrahedral intermediate of step 1 and step 3 are generated, the negative oxygen ion, having accepted the electrons from the carbonyl double bond fits perfectly into the oxyanion hole. In effect, serine proteases preferentially bind the transition state and the overall structure is favored, lowering the activation energy of the reaction. This "preferential binding" is responsible for much of the catalytic efficiency of the enzyme.